Missing Higher Order Uncertainties with focus on top-antitop production

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Uncertainties in precision phenomenology

Experiments are getting more precise → theory uncertainties matter!

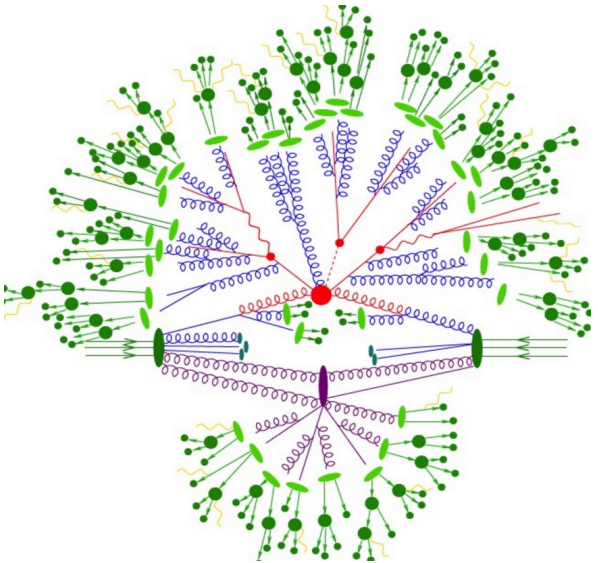
Sources of theory uncertainties:

- parametric (values of coupling parameters etc.)
→ variation of parameters within their uncertainties

- parton distribution functions (PDFs)
→ different error propagation methods (fit parameter, replicas,...)

- non-perturbative parameters in Monte Carlo simulations.
→ needs data constraints by definition. Problematic if dominant effect...

- missing higher orders in fixed-order and resummed predictions (MHOU)
→ tricky because we are trying to estimate the unknown….



[Credit: SHERPA]

Missing higher orders

Notation from: [Tackmann 2411.18606]
Beyond Scale Variations: Perturbative Theory Uncertainties from Nuisance Parameters

Generic perturbative expansion:

$$f(\alpha) = f_0 + f_1\alpha + f_2\alpha^2 + f_3\alpha^3 + \dots$$

f_i : the coefficient of the series, potentially unknown

We can compute the truncated series: $\hat{f}_i$ → the true value, i.e. a value we actually computed

$$f^{\mathrm{LO}}(\alpha) = \hat{f}_0 \qquad f^{\mathrm{NLO}}(\alpha) = \hat{f}_0 + \hat{f}_1\alpha \qquad f^{\mathrm{NNLO}}(\alpha) = \hat{f}_0 + \hat{f}_1\alpha + \hat{f}_2\alpha^2$$

The **missing terms are the source of uncertainty**.
(assume convergence → the first missing is the dominant one)

$$f^{\mathrm{LO+1}}(\alpha) = \hat{f}_0 + f_1\alpha \quad f^{\mathrm{NLO+1}}(\alpha) = \hat{f}_0 + \hat{f}_1\alpha + f_2\alpha^2 \quad f^{\mathrm{NNLO+1}}(\alpha) = \hat{f}_0 + \hat{f}_1\alpha + \hat{f}_2\alpha^2 + f_3\alpha^3$$

Challenge: how to estimate f_1, f_2, f_3, … without computing them?

Theory uncertainties from scale variations

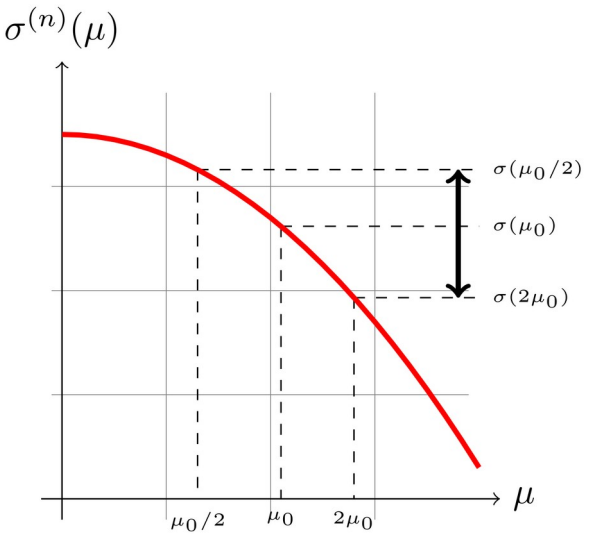
Lets focus on QCD as an example: $\alpha = \alpha_s(\mu_0)$

$$\mathrm{d}\sigma = \mathrm{d}\sigma^{(0)} + \alpha_s \mathrm{d}\sigma^{(1)} + \alpha_s^2 \mathrm{d}\sigma^{(2)} + \ldots \xrightarrow{\text{RGE}} \mu \frac{\mathrm{d}\sigma^{(n)}}{\mathrm{d}\mu} = \mathcal{O}\left(\alpha_s^{(n_0+n+1)}\right)$$

Of same order as the next dominant term → exploiting this to estimate size of $\mathrm{d}\sigma^{(n+1)}$

Scale variation prescription (ad-hoc and heuristic choice)

- choose 'sensible' μ_0
  - → principle of fasted apparent convergence: $\sigma^{(n)}(\mu_{\mathrm{FAC}}) = 0$
  - → principle of minimal sensitivity $\mu \frac{\mathrm{d}\sigma(\mu)}{\mathrm{d}\mu}\Big|_{\mu=\mu_{\mathrm{PMC}}} = 0$
  - → ...
- vary with a factor (typically 2)
- take envelope as uncertainty



Scale variation approach

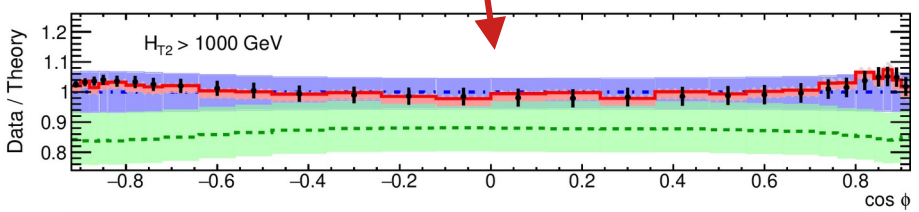
Estimates from scale variations for the unknown f_i:

$$f_2 \equiv \Delta f^{\mathrm{NLO}} = f^{\mathrm{NLO}} - \tilde{f}^{\mathrm{NLO}}(\tilde{\alpha}) = -\alpha^2 b_0 \hat{f}_1 + \mathcal{O}(\alpha^3)$$

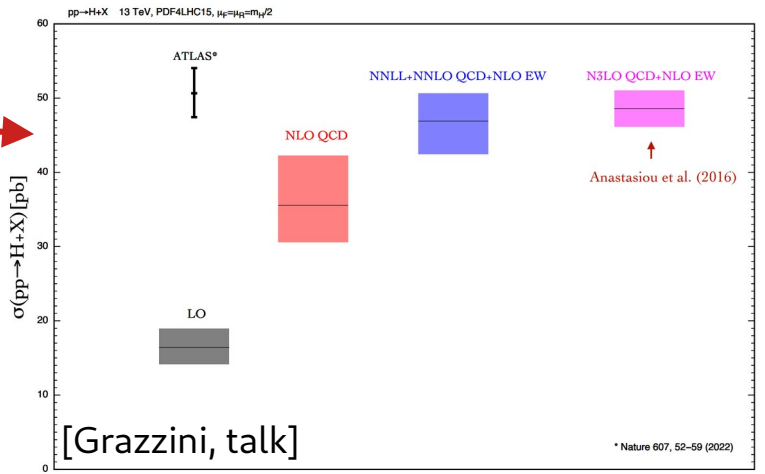
$$f_3 \equiv \Delta f^{\mathrm{NNLO}} = f^{\mathrm{NNLO}} - \tilde{f}^{\mathrm{NNLO}}(\tilde{\alpha}) = \alpha^3 (2b_0(\hat{f}_2 - b_0 \hat{f}_1) + b_1 \hat{f}_1) + \mathcal{O}(\alpha^4)$$

$$b_0 = \frac{\beta_0}{2\pi} L \qquad b_1 = \frac{\beta_0^2}{4\pi^2} L^2 + \frac{\beta_1}{8\pi^2} L \qquad b_2 = \frac{\beta_0^3}{8\pi^3} L^3 + \frac{5\beta_0\beta_1}{32\pi^2} L^2 + \frac{\beta_2}{32\pi^3} L \qquad L = \ln \frac{\mu_0}{\mu}$$

Sometimes it does works … sometimes not



[ATLAS 2301.09351]



[Grazzini, talk]

Short comings of scale variations

- not always reliable … however in most cases issues are understood/expected:
  new channels, phase space constraints, etc. → often we can design workarounds
- however, some issues are more fundamental:
  - → how to choose the central scale? → not a physical parameter, no 'true' value
    (Principle of fasted apparent convergence, principle of minimal sensitivity,…)
  - → how to propagate the estimated uncertainty, no statistical interpretation!
  - → what about correlations? Based on 'fixed form' of the lower orders and RGE.
- Alternative approaches:

"Bayesian"

[Cacciari,Houdeau 1105.5152]
[Bonvini 2006.16293]
[Duhr, Huss, Mazeliauskas, Szafron 2106.04585]

"Theory Nuisance Parameter"

[Tackmann 2411.18606]
→ W mass extraction: [CMS 2412.13872]
[Cal,Lim, Scott,Tackmann Waalewijn 2408.13301]
[Lim, Poncelet, 2412.14910]
[Cridge,Marinelli,Tackmann,2506.13874]

Introducing theory nuisance parameters (TNPs)

[Tackman 2411.18606]

Generic perturbative expansion:

$$f(\alpha) = f_0 + f_1\alpha + f_2\alpha^2 + f_3\alpha^3 + \dots$$

$$f^{\mathrm{LO+1}}(\alpha) = \hat{f}_0 + f_1(\theta)\alpha$$

$$f^{\mathrm{NLO+1}}(\alpha) = \hat{f}_0 + \hat{f}_1\alpha + f_2(\theta)\alpha^2$$

$$f^{\mathrm{NNLO+1}}(\alpha) = \hat{f}_0 + \hat{f}_1\alpha + \hat{f}_2\alpha^2 + f_3(\theta)\alpha^3$$

Not knowing these constitute the (leading) source of MHOU

Introduce a parametrisation of unknown coefficients in terms of

"Theory nuisance parameters" θ

Key features

1) The parametrization such that there is a true value: $f_i(\hat{\theta}) = \hat{f}_i$

2) Distributions of θ "known" (for example from already existing computations)

→ "Expert knowledge" to construct such a parametrisation

Finding a parametrization

How to find a parametrization for $f_i(\theta)$? What if it depends on an additional parameter x?

- it is a constant → do I understand its normalization?
- it is function of x, but we know the overall functional dependence, e.g.

$$f_i(x,\theta) = \sum_k^{m_i} c_i^k(\theta)\ln^k(x)$$

- it is a function of x, but don't really know the functional dependence

$$f_i(x,\theta) = \sum_k^{\infty} c_i^k(\theta)\phi^k(x)$$ ← Some basis of functions: Legendre, Chebyshev,...

If this series converges: $f_i(x,\theta) \approx \sum_k^{m} c_i^k(\theta)\phi^k(x)$

Finding a parametrization

How to find a parametrization for $f_i(\theta)$? What if it depends on an additional parameter x?

- Multi-dimensional dependences: $f_i(x,y,\theta) = \sum_{k,l} c_i^{kl}(\theta) f^{kl}(x,y)$ ...
- Often low-dimensional observables
  → y integrated out: $f_i(x,\theta) = \int \mathrm{d}y f_i(x,y,\theta)$

  if it factorizes $f_i(x,y,\theta) = h_i(x,\theta) g_i(y,\theta)$

  for example: $\int \mathrm{d}y f_i(x,y,\theta) = h_i(x,\theta) G_i(\theta) \approx h_i(x,\theta) \tilde{G}_i \theta$

To summarize: even if you don't fully know the form of f_i you can systematically build it up
→ what are the dominant effects?
→ what dependencies/correlations are actually important?

Example: TNPs in resummed cross sections

[Tackman 2411.18606]

Transverse momentum resummation at leading power ($\mu = Q$):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}p_T} = [\underbrace{H \times B_a \times B_b \times S}](\alpha_s, \ln\left(\frac{p_T}{Q}\right)) + \mathcal{O}\left(\frac{p_T^2}{Q^2}\right)$$

+ non pert.
+ quark mass
+ QED/EW

$$X(\alpha_s, L) = X(\alpha_s) \exp \int_0^L \mathrm{d}L' \{\Gamma(\alpha_s(L'))L' + \gamma_X(\alpha_s(L'))\}$$

Perturbative expansion of resummation ingredients:
→ functional dependence of H and S on ln(pT/Q) determined through RGE
→ only needs boundary conditions/anomalous dimensions:

$$X(\alpha_s) = X_0 + \alpha_s X_1 + \alpha_s^2 X_2 + \cdots + \alpha_s^n X_n(\theta)$$

$$\Gamma(\alpha_s) = \alpha_s[\Gamma_0 + \alpha_s \Gamma_1 + \alpha_s^2 \Gamma_2 + \cdots + \alpha_s^n \Gamma_n(\theta)]$$

$$\gamma(\alpha_s) = \alpha_s[\gamma_0 + \alpha_s \gamma_1 + \alpha_s^2 \gamma_2 + \cdots + \alpha_s^n \gamma_n(\theta)]$$

Task: find suitable parametrization and variation range

These are numbers here
→ only need normalisation

Example: TNPs in resummed cross sections

[Tackman 2411.18606]

Transverse momentum resummation at leading power ($\mu = Q$):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}p_T} = [\underbrace{H \times B_a \times B_b \times S}](\alpha_s, \ln\left(\frac{p_T}{Q}\right)) + \mathcal{O}\left(\frac{p_T^2}{Q^2}\right)$$

+ non pert.
+ quark mass
+ QED/EW

$$X(\alpha_s, L) = X(\alpha_s) \exp \int_0^L \mathrm{d}L' \{\Gamma(\alpha_s(L'))L' + \gamma_X(\alpha_s(L'))\}$$

Perturbative expansion of resummation ingredients:

→ beam functions B are bit trickier:

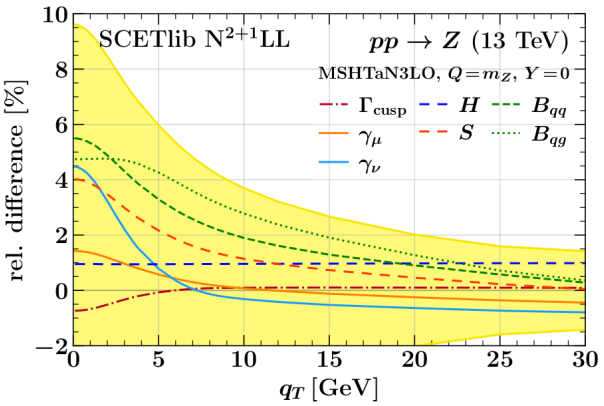
→ x – dependence determined by PDFs and matching kernels (eff. splitting functions)

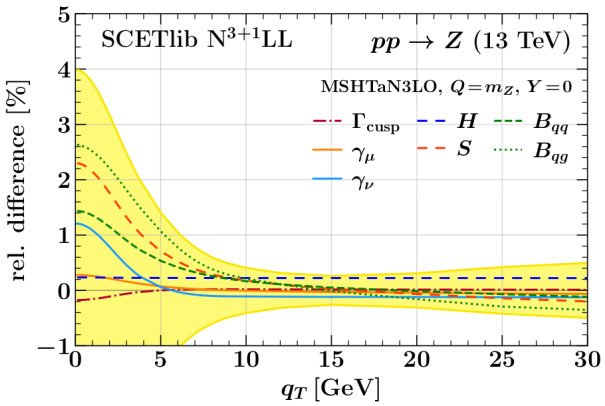
$$\tilde{b}_{i,n}(x) = \sum_j \int \frac{\mathrm{d}z}{z} \tilde{I}_{ij,n}(z)\, f_j\left(\frac{x}{z}\right)$$

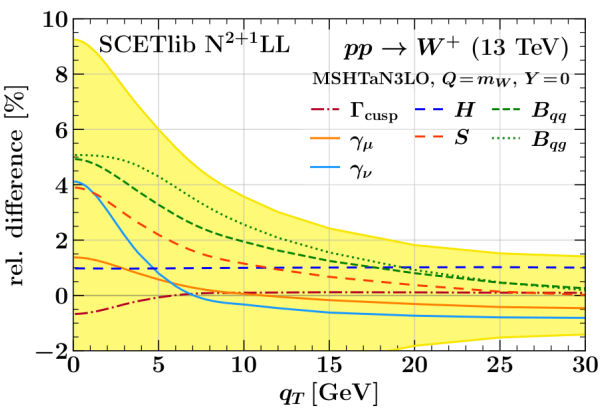
→ if we don't need correct correlations in the boson rapidity → simplified ansatz:

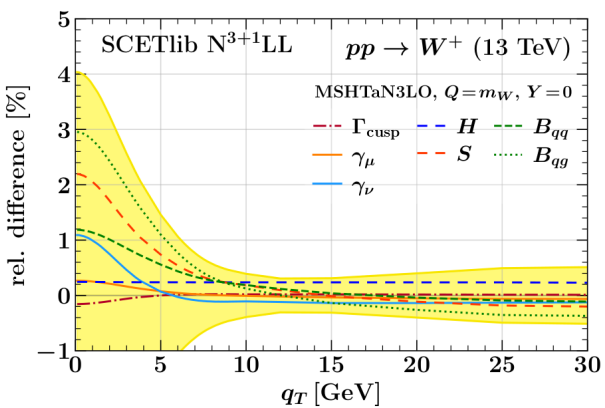
$$\tilde{I}_{ij,n}(z, \theta_n^{B_{ij}}) = \frac{3}{2}\, \theta_n^{B_{ij}}\, \hat{\tilde{I}}_{ij,n}(z)$$

Point-wise correlations from TNP variations









Lines: individual +- 1 variation

Yellow band:
quadrature sum of all components (sym)

Example 2: small-R resummation

Semi-inclusive jet function [1606.06732, 2410.01902]

$$\frac{d\sigma_{\rm LP}}{dp_T d\eta} = \sum_{i,j,k} \int_{x_{i,\min}}^1 \frac{dx_i}{x_i} f_{i/P}(x_i,\mu) \int_{x_{j,\min}}^1 \frac{dx_j}{x_j} f_{j/P}(x_j,\mu) \int_{z_{\min}}^1 \frac{dz}{z} \mathcal{H}_{ij}^k(x_i,x_j,p_T/z,\eta,\mu) \times J_k\left(z, \ln\frac{p_T^2 R^2}{z^2\mu^2},\mu\right),$$

The same hard function as for identified hadrons!

Modified RGE:

[2402.05170,2410.01902]

$$\frac{d\vec{J}\left(z, \ln\frac{p_T^2R^2}{z^2\mu^2},\mu\right)}{d\ln\mu^2} = \int_z^1 \frac{dy}{y} \vec{J}\left(\frac{z}{y}, \ln\frac{y^2 p_T^2 R^2}{z^2\mu^2},\mu\right) \cdot \widehat{P}_T(y)$$

TNPs for inclusive jet production

Thanks to Terry Generet to put this together! [2606.xxxx]

The pT spectrum is a steeply falling function → effectively only few Mellin moments contribute

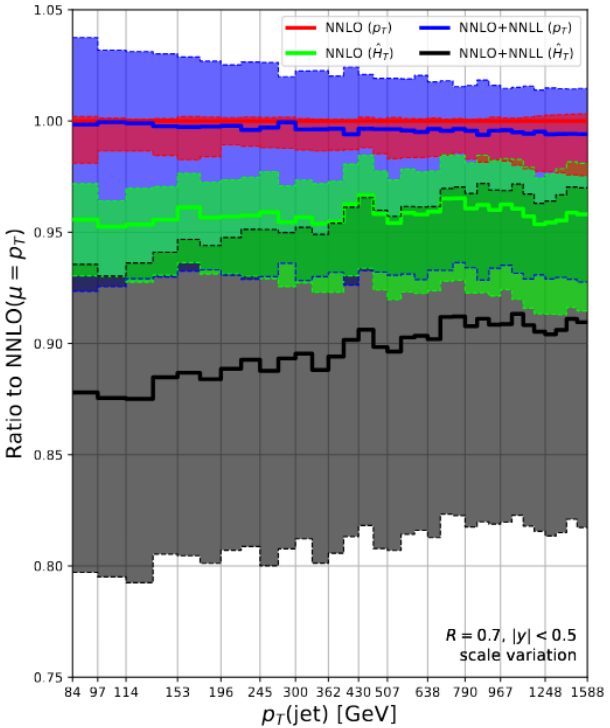
$$\frac{d\sigma}{dp_T} \approx \sum_{a,b} L_{ab}(\hat{E}/E = 2p_T/E)\frac{d\hat{\sigma}_{ab}}{dp_T}(N = \tilde{n}(2p_T/E))$$

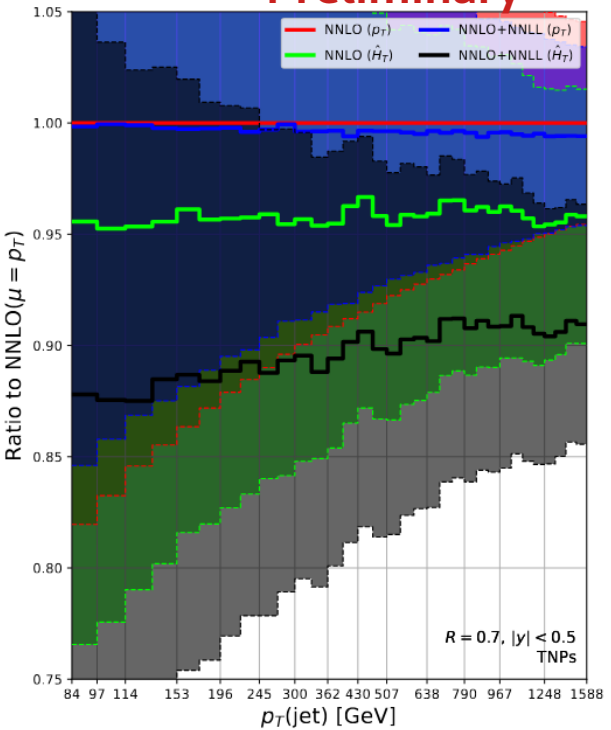
$$\begin{aligned} d\hat{\sigma}_{ab\to cd}(N) = {} & J_{\text{in}}^{(a)}\left(\frac{\hat{s}}{N_{0a}^2\mu^2}, \alpha_s(\mu)\right) J_{\text{in}}^{(b)}\left(\frac{\hat{s}}{N_{0b}^2\mu^2}, \alpha_s(\mu)\right) \\ & \times J_{\text{fr}}^{(c)}\left(\frac{\hat{s}}{N_0^2\mu^2}, \alpha_s(\mu)\right) J_{\text{rec}}^{(d)}\left(\frac{\hat{s}}{N_0\mu^2}, \frac{\hat{s}}{N_0^2\mu^2}, \alpha_s(\mu)\right) \\ & \times \text{Tr}\left[\mathbf{H}_{ab\to cd}\left(\frac{\hat{s}}{\mu^2}, \alpha_s(\mu)\right)\mathbf{S}_{ab\to cd}\left(\frac{\hat{s}}{N_0^2\mu^2}, \alpha_s(\mu)\right)\right] + \mathcal{O}\left(\frac{1}{N}\right) \end{aligned}$$

These then can be broken down into scalar series:
(soft+hard functions require approx. of colour matrix → error on the error)

$$\begin{aligned} J_{\text{in}}^{(i)}\left(\frac{\hat{s}}{N_{0i}^2\mu^2}, \alpha_s(\mu)\right) = J_{\text{fr}}^{(i)}\left(\frac{\hat{s}}{N_{0i}^2\mu^2}, \alpha_s(\mu)\right) = R_i\big(\alpha_s(\mu)\big) \\ \times \exp\left[\int_{\sqrt{\hat{s}}/N_{0i}}^{\mu}\frac{d\mu'}{\mu'}\left(A_i\big(\alpha_s(\mu')\big)\ln\left(\frac{\mu'^2 N_{0i}^2}{\hat{s}}\right) - \frac{1}{2}D_i\big(\alpha_s(\mu')\big)\right)\right] \end{aligned}$$

Example 2: absolute results

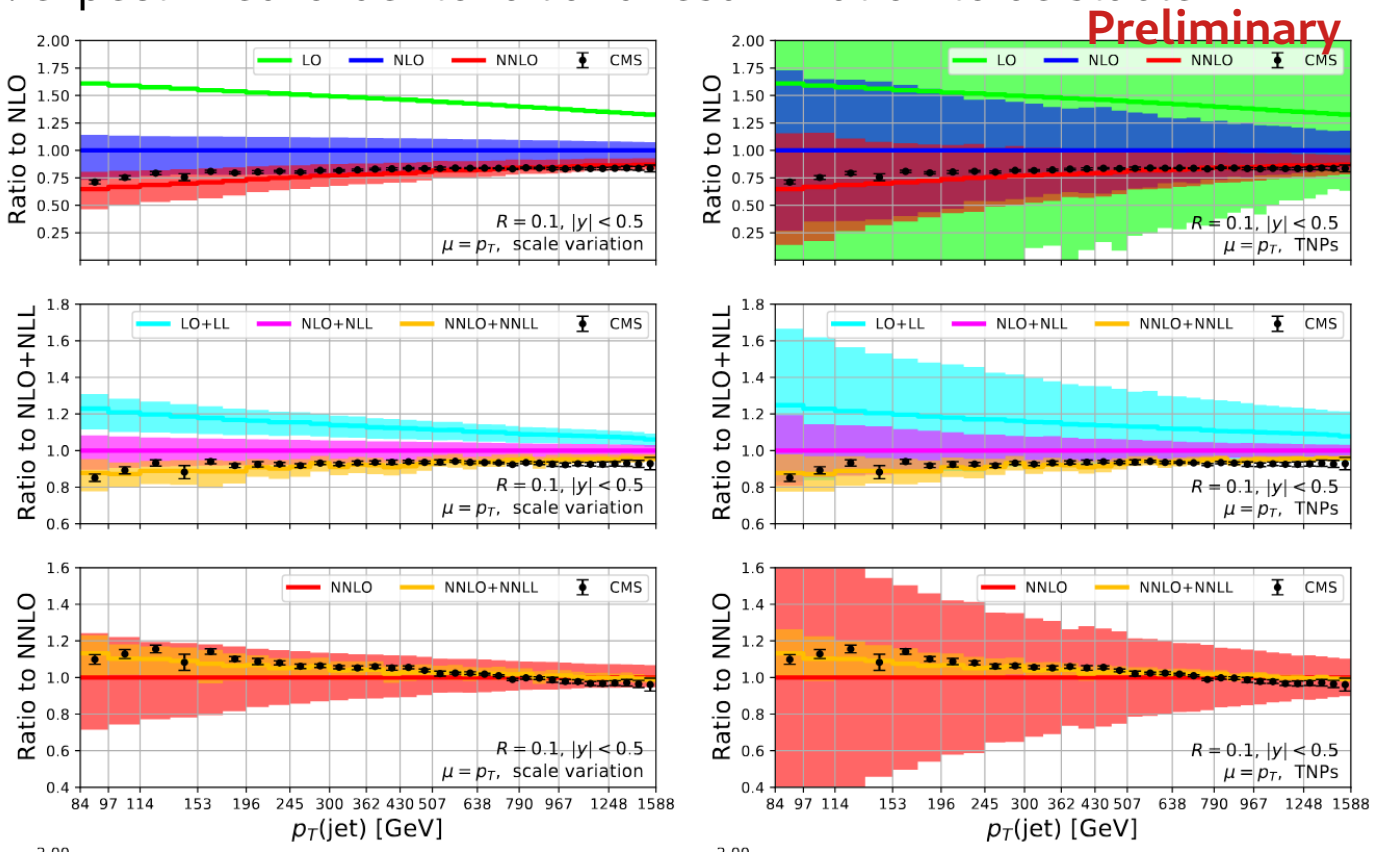




- TNPs imply larger uncert.
  scale: 2-3% FO, 5-10% res.
  TNP: ~ 15% (low pT), 5% (high pT)
- Much stronger PT dependence of uncertainty for TNP
- taking the spread of predictions as gauge
  (scales +/- res.)
  → TNPs cover it
  → scale var. doesn't

Example 2: ratios of different radii

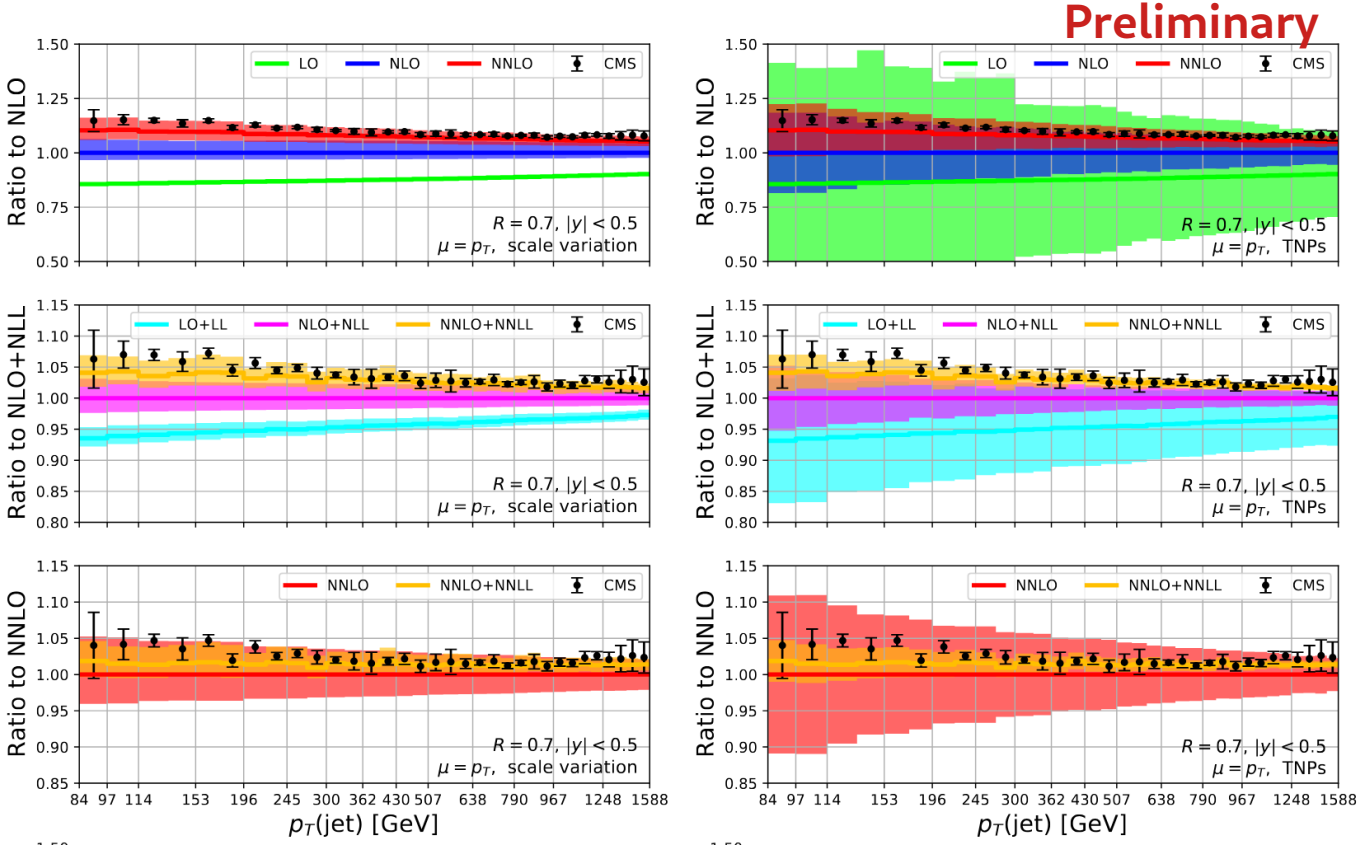
Small R: expect fixed-order to fail and resummation to be stable



side note
these are ratios (R/R=0.4), TNPs allow **correct correlation**!

Example 2: ratios of different radii

Intermediate R: observed small scale dependence → TNPs more realistic



Take-home message
TNP unc. models for 'complictated' observables can be done

Some thoughts…

Picture that emerges from the application of TNPs to resummed x-sections:
simple ingredients that enter different computations/processes etc. → ideal situation

But actually not that simple:

- Scheme dependence of ingredients (canonical choices make life easier)?
  e.g. scales, IR subtraction, … → needs modified parametrisations
- Some TNPs represent directly numbers: $\Gamma,\ \gamma,\ H$ for simple processes
  but others are functions → Beam functions, hard functions for more complicated processes
  → requires in practise (converging) approximations/expansions
- "Easy to implement" for use-cases so far (effectively analytic resummation → quite fast)
  → might be really expensive if each variation needs a expensive computation (Monte Carlos,…)

What can you do if you have only little knowledge?

TNP approach for fixed-order computations

[Lim, Poncelet, 2412.14910]

- it is a function of x, but don't really know the functional dependence

$$f_i(x,\theta) = \sum_k^{\infty} c_i^k(\theta)\phi^k(x)$$ ← Some basis of functions: Legendre, Chebyshev,...

If this series converges: $f_i(x,\theta) \approx \sum_k^{m} c_i^k(\theta)\phi^k(x)$

$$\begin{aligned}\mathrm{d}\sigma &= \alpha_s^n N_c^m \,\mathrm{d}\bar{\sigma}^{(0)} + \alpha_s^{n+1} N_c^{m+1} \,\mathrm{d}\bar{\sigma}^{(1)} + \alpha_s^{n+2} N_c^{m+2} \,\mathrm{d}\bar{\sigma}^{(2)} + \dots \\ &= \alpha_s^n N_c^m \,\mathrm{d}\bar{\sigma}^{(0)} \left[1 + \alpha_s N_c \left(\frac{\mathrm{d}\bar{\sigma}^{(1)}}{\mathrm{d}\bar{\sigma}^{(0)}}\right) + \alpha_s^2 N_c^2 \left(\frac{\mathrm{d}\bar{\sigma}^{(2)}}{\mathrm{d}\bar{\sigma}^{(0)}}\right) + \dots\right]\end{aligned}$$

Observation for QCD corrections: $\dfrac{\mathrm{d}\bar{\sigma}^{(i)}}{\mathrm{d}\bar{\sigma}^{(0)}} \sim \mathcal{O}(1)$

Use some knowledge about lower orders but introduce parametric dependence:

$$\frac{\mathrm{d}\bar{\sigma}_{\mathrm{TNP}}^{(N+1)}}{\mathrm{d}\bar{\sigma}^{(0)}} = \sum_{j=1}^{N} f_k^{(j)}\left(\vec{\theta}, x\right)\left(\frac{\mathrm{d}\bar{\sigma}^{(j)}}{\mathrm{d}\bar{\sigma}^{(0)}}\right)$$

Chebyshev: $f_k^C\left(\vec{\theta}, x\right) = \dfrac{1}{2}\sum_{i=0}^{k} \theta_i T_i(x)$

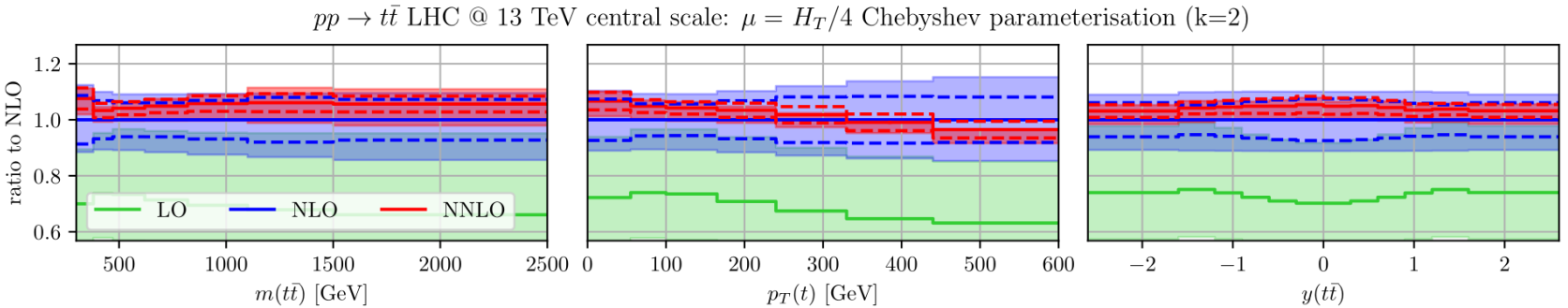
$x \in [-1, 1]$

Truncated exp. → there is only $f_i(\hat{\theta}) \approx \hat{f}_i$
→ correspond to the extreme case with all other dependencies are integrated out

See also MSHT [2207.04739]

Application to ttbar

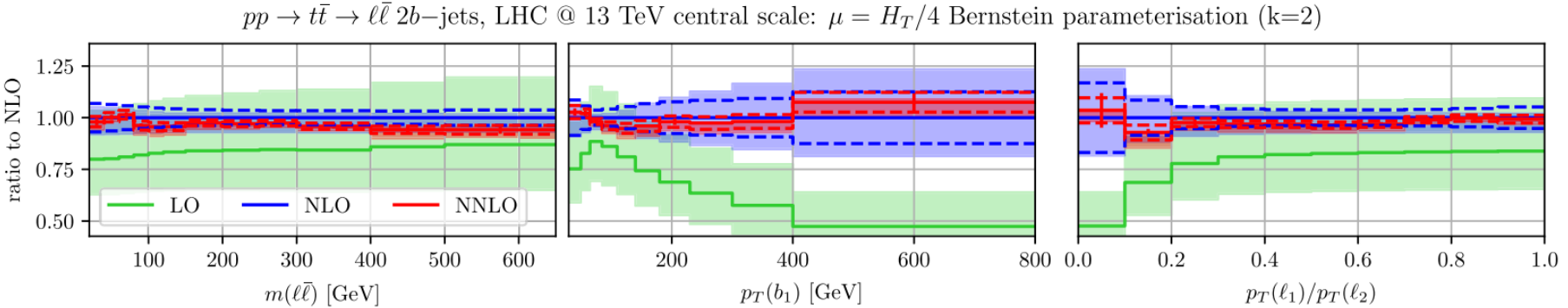
Stable tops



Solid – 7pt scale var

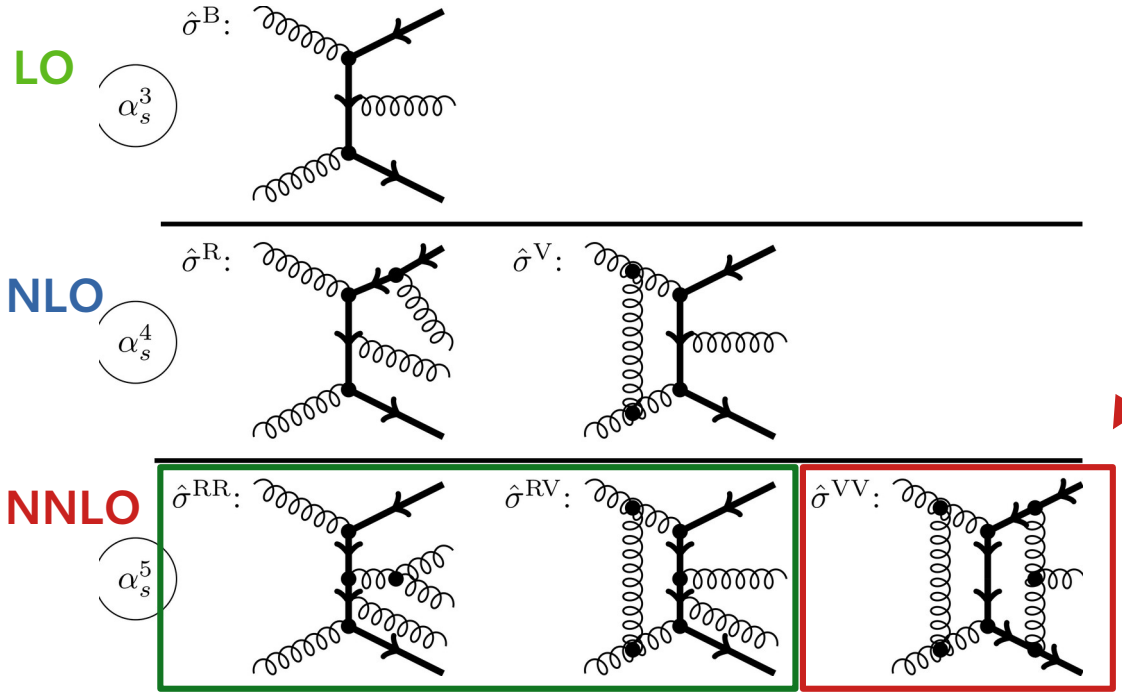
Dashed - TNPs 65%cl

Tops + NWA decays



ttj through NNLO QCD

Higher-order QCD corrections to top-quark pair production in association with a jet, Badger, Becchetti, Brancaccio, Czakon, Hartanto, Poncelet, Zoia [arxiv:2511.11431]



Complete colour dependence

Leading color approximation for two-loop finite remainder

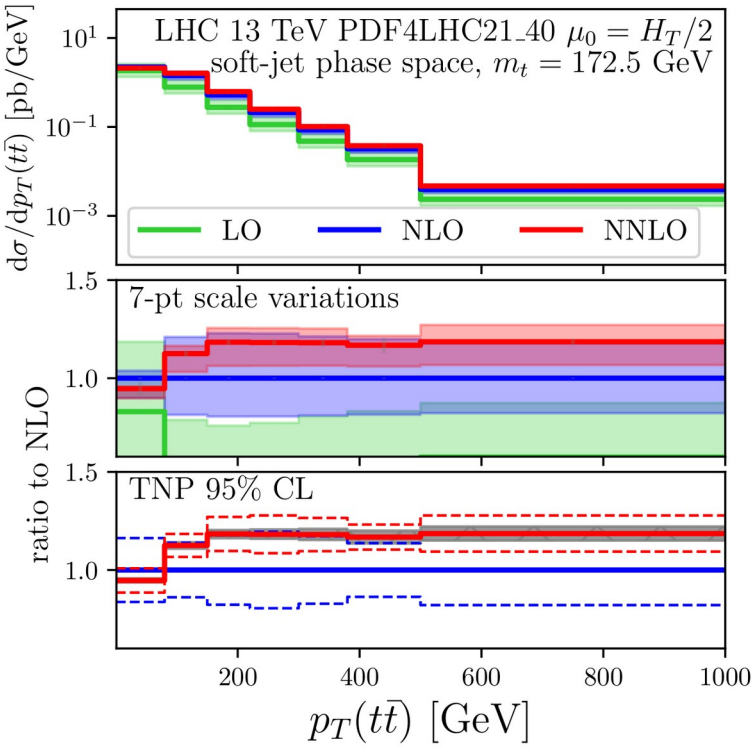
Double virtual QCD corrections to $t\bar{t}$+jet production at the LHC, Badger, Becchetti, Brancaccio, Czakon, Hartanto, **Poncelet**, Zoia [arxiv:2511.11424]

Log-dependence (only lower loops) in **full color**

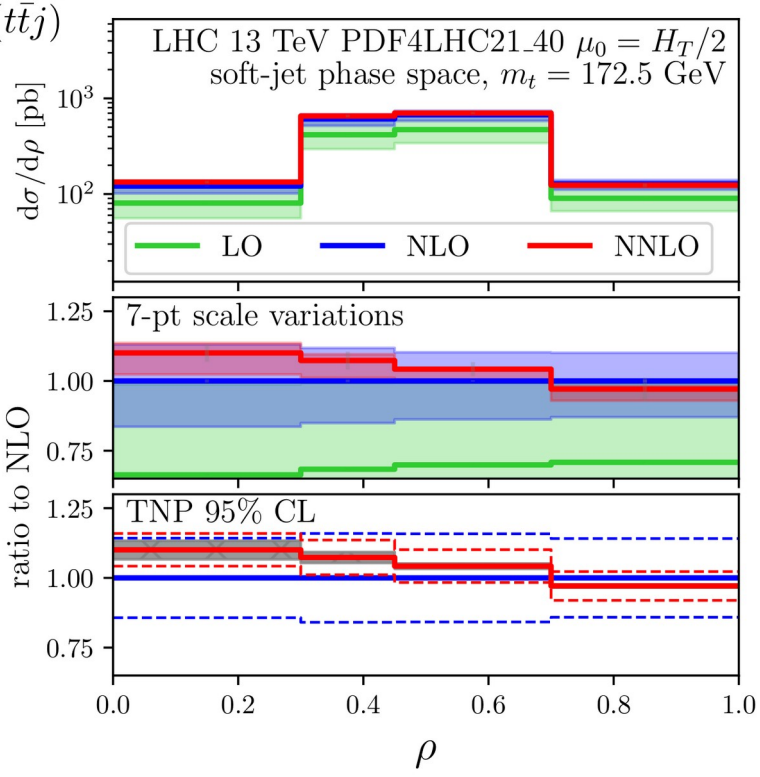
$$F^{(2)}_{\text{l.c. resc.}}(\mu^2) = \frac{F^{(0)}}{F^{(0)}_{\text{l.c.}}} F^{(2)}_{\text{l.c.}}(Q^2) + \sum_{i=0}^{4} c_i \log^i(\mu^2/Q^2)$$

Rescaling captures some of the effect

ttj: TNP uncertainty estimates



$$\rho = \frac{2m_0}{m(t\bar{t}j)}$$



… in summary good agreement between ‘effective’ TNP and scale variations for top-quarks.

ttj ‘sub-leading colour’ uncertainty

Full NNLO QCD except for two-loop virtuals: $$F^{(2)}_{\text{l.c. resc.}}(\mu^2) = \frac{F^{(0)}}{F^{(0)}_{\text{l.c.}}} F^{(2)}_{\text{l.c.}}(Q^2) + \sum_{i=0}^{4} c_i \log^i(\mu^2/Q^2)$$

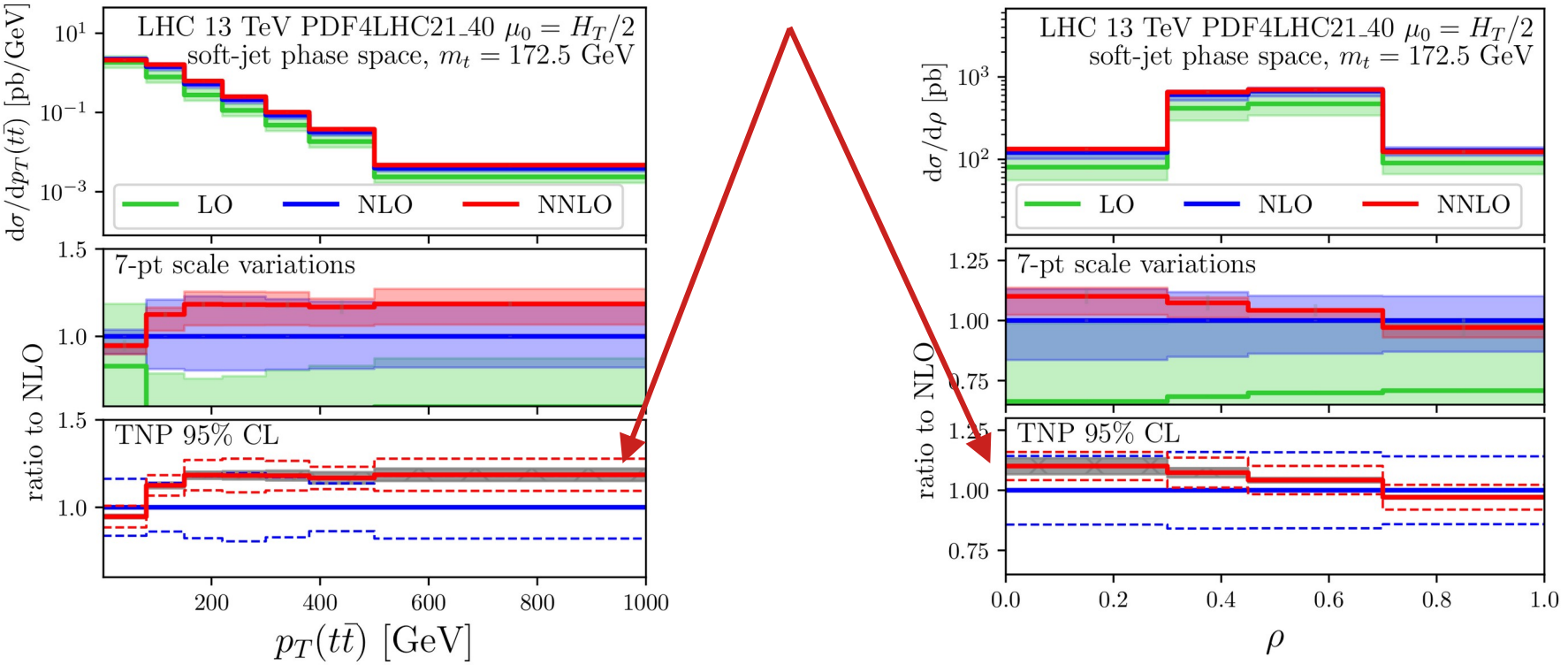
Naive ‘TNP‘-parametrisation of sub-leading colour: $$\frac{F^{(2)}_{l.c.}}{F^{(0)}_{l.c.}} \to \frac{F^{(2)}_{l.c.}}{F^{(0)}_{l.c.}} + \frac{\theta}{N_c^2} \frac{F^{(2)}_{l.c.}}{F^{(0)}_{l.c.}} \qquad \theta \in [-1, 1]$$

Have also a look for something similar in 2→ 3 jet/photon processes:

How much color do we really need? Two-loop subleading-color effects in photon and jet physics [Czakon, Poncelet 2512.17591]

ttj: subleading color estimates

Sub-leading colour unc. → grey band



Caveat: might fail close to threshold due to octet contributions! → needs some refinement

Take home message

- Increasing precision demands accurate theory uncertainty estimates
- De-facto standard: scale variations
  → various short-comings: robustness, no statistical interpretation, correlations,...
- Alternative approaches to scale variations: Bayesian and TNP approach
- **Theory Nuisance Parameters**
  - In principle less biased, better correlations → does not depend on any “known” orders
  - Allows for a statistical interpretation and constraints from data!
  - Fixed-order tricky, not much knowledge about higher-order terms
    → fallback to general parameterization

    If factorization formula available → feasible but observable dependent!
    TNP for jet cross sections show promising first results

Backup

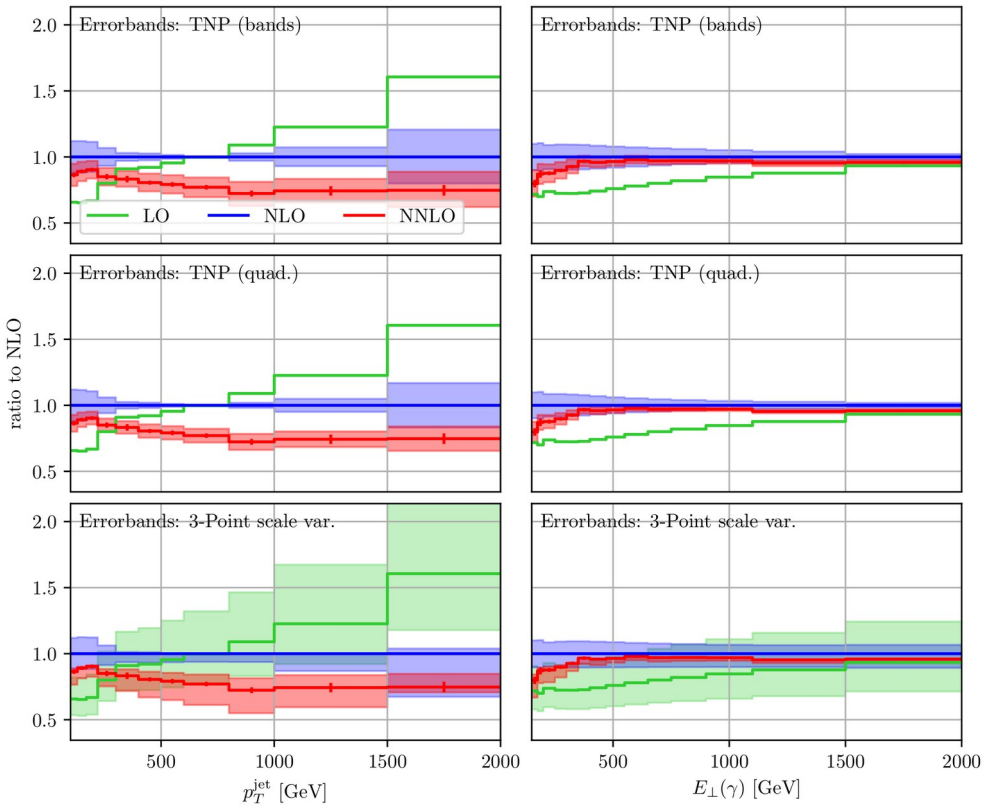
Points for discussion, caveats and open questions

Some arising questions regarding fixed-order model: $\frac{\mathrm{d}\bar{\sigma}^{(N+1)}_{\mathrm{TNP}}}{\mathrm{d}\bar{\sigma}^{(0)}} = \sum_{j=1}^{N} f_k^{(j)}\left(\vec{\theta}, x\right)\left(\frac{\mathrm{d}\bar{\sigma}^{(j)}}{\mathrm{d}\bar{\sigma}^{(0)}}\right)$

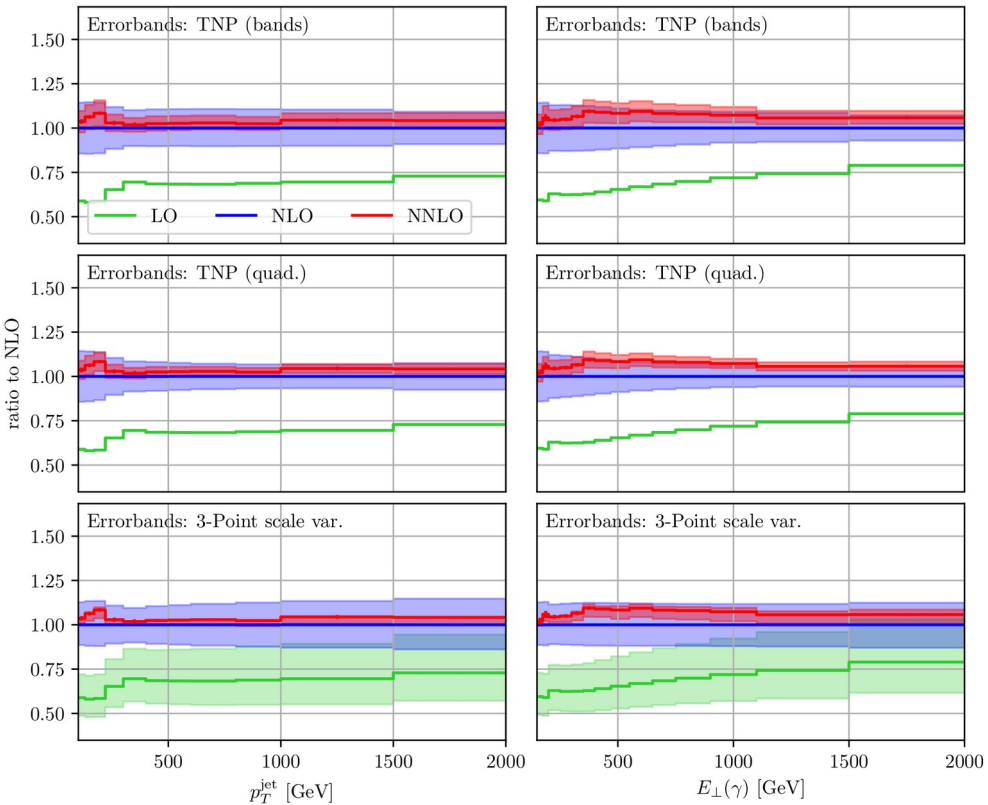
- How does the uncertainty estimate depend on the central scale choice?
  → bad scale choices lead to large uncertainties by construction due to large corrections.
- What about NLO uncertainty if $\mathrm{d}\bar{\sigma}^{(1)} = 0$ for given scale?
  → amend parametrisation by j = 0 term.
- Each parametrisation is for one observable at a time:
  How to deal with higher dimensional distributions? Consistency upon integration?
  → WIP
- What about EW corrections?
  → Sudakov logs should work well! → Radiation from resonances more difficult.
- **How to correlate different processes at fixed-order?**
  → ???, would require something like: $\mathrm{d}\bar{\sigma}^{(n)}(\theta) = \mathrm{d}\Phi \left\langle M^0 \right| \mathcal{P}(\theta) \left| M^0 \right\rangle$ $\mathcal{P}(\theta)$ → process-independent "operator"
- How sensitive are we to the parametrisation? How many terms?
  → two quite general parametrisations tested, increase degree by demand.

Challenging case

"Bad" scale choice $\mu = E_T$ no j=0 term

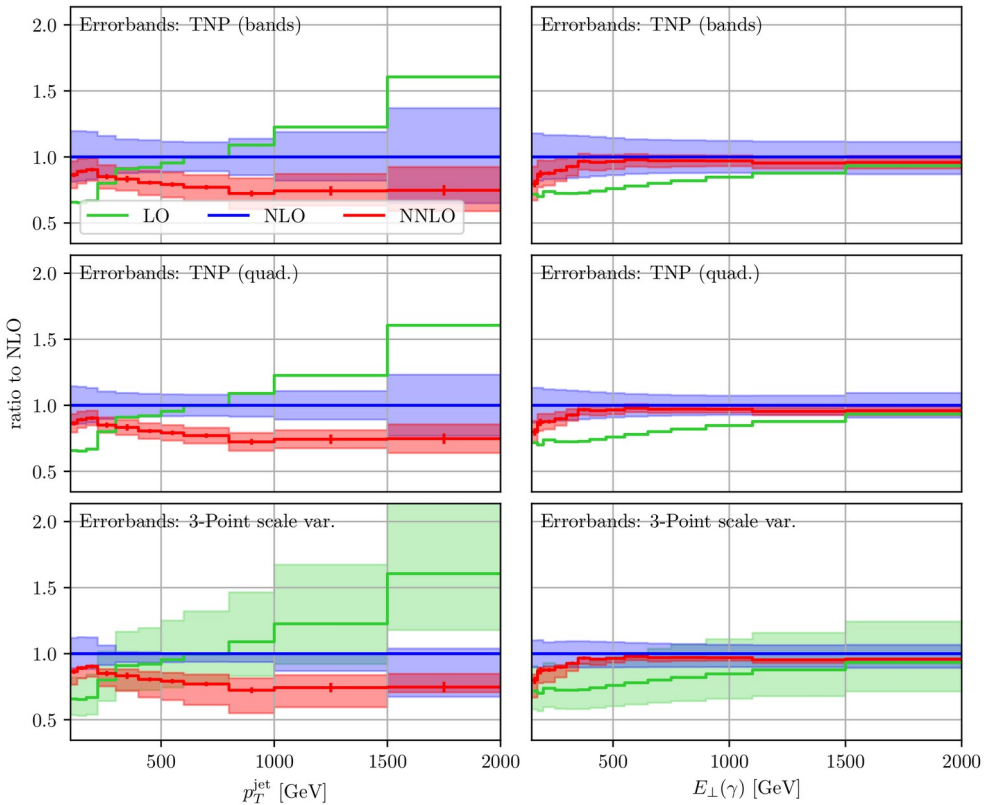


"Good" choice $\mu = H_T$ no j=0 term

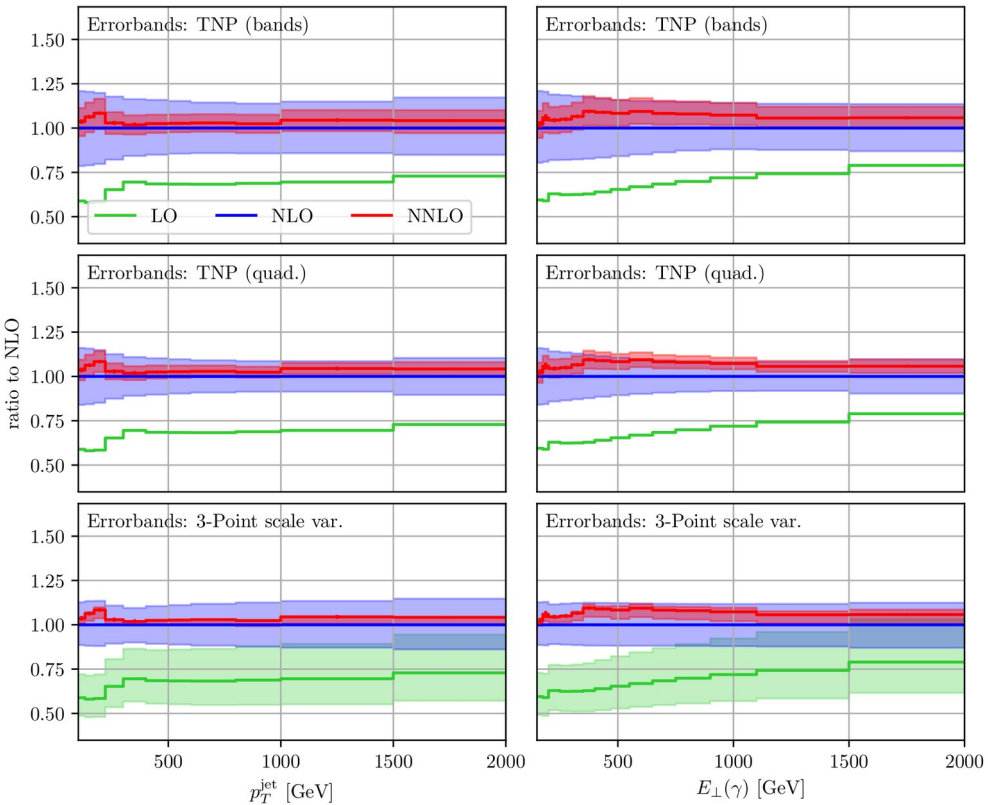


Challenging case → extended parametrisations

"Bad" scale choice $\mu = E_T$ with j=0 term

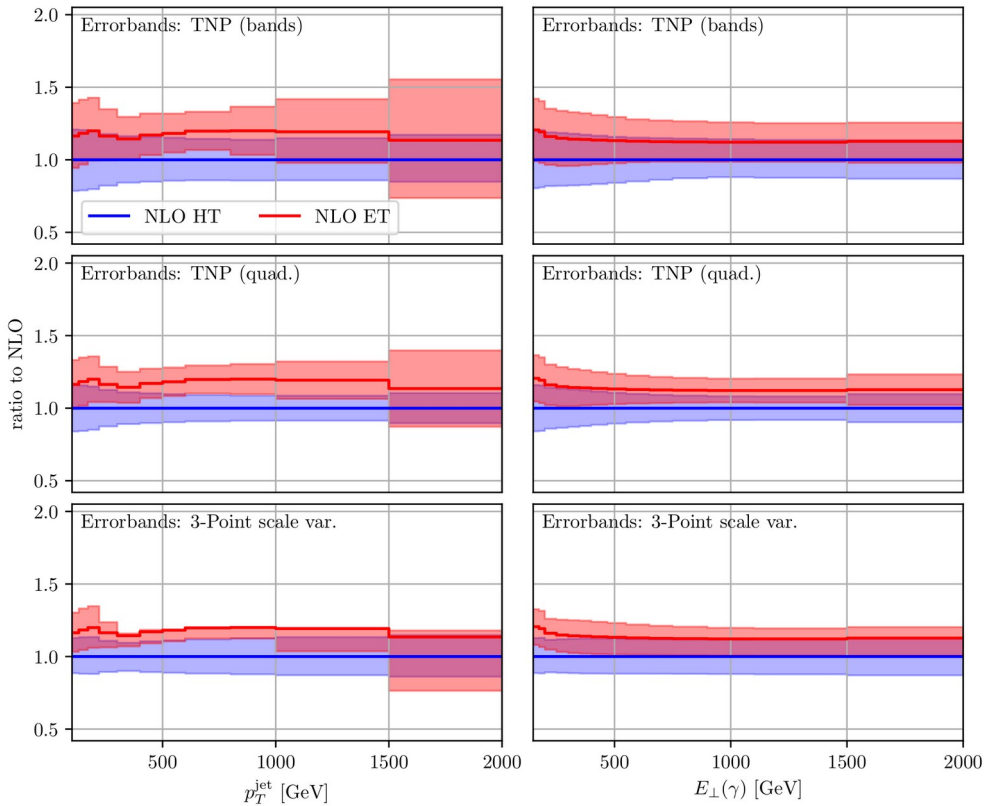


"Good" choice $\mu = H_T$ with j=0 term

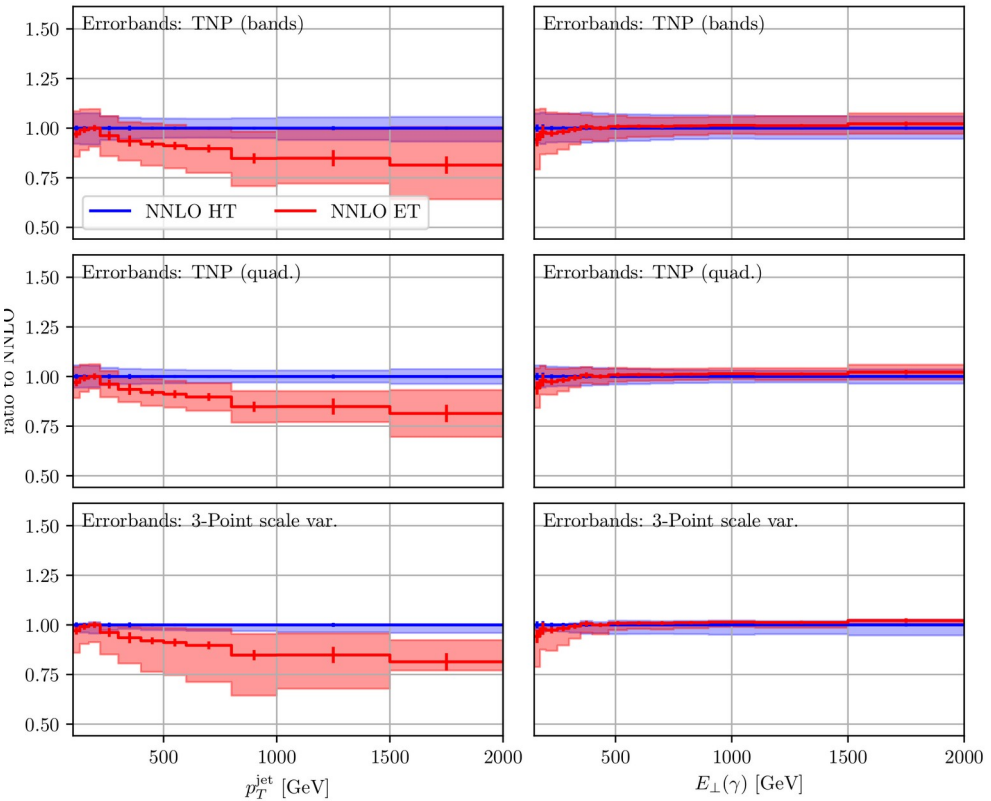


Challenging case → comparisons

NLO QCD



NNLO QCD



Bayesian approach I

→ Instead of ad-hoc fixed variation try to give some probabilistic interpretation

$$\mathrm{d}\sigma = \mathrm{d}\sigma^{(0)}(1 + \delta^{(1)} + \delta^{(2)} + \dots)$$

Probability to find coefficient $\delta^{(n+1)}$ given $\delta^{(n)}$: [Cacciari,Houdeau 1105.5152]

$$P(\delta^{(n+1)}|\delta^{(n)}) = \frac{P(\delta^{(n+1)})}{P(\delta^{(n)})} = \frac{\int \mathrm{d}a P(\delta^{(n+1)}|a) P_0(a)}{\int \mathrm{d}a P(\delta^{(n)}|a) P_0(a)}$$

Need to provide model and prior

Bayes: $P(A|B) = P(B|A)P(A)/P(B)$ with: $P(\delta^{(n)}|\delta^{(n+1)}) = 1$

CH model: $\delta_k = c_k \alpha_s^k$ c_k come from geometric series: $|c_k| \leq \overline{c} \quad \forall k$

Geometric model: $|\delta_k| \leq c a^k \quad \forall k$ [Bonvini 2006.16293]

abc model: $b - c \leq \frac{\delta_k}{a^k} \leq b + c \quad \forall k$ [Duhr, Huss, Mazeliauskas, Szafron 2106.04585]

Bayesian approach II

Inclusion of scale dependence: $$P(\delta_{n+1}|\delta_n) = \int \mathrm{d}\mu P(\delta_{n+1}|\delta_n;\mu)P(\mu|\delta_n)$$

Scale marginalisation (the scale becomes a model parameter)

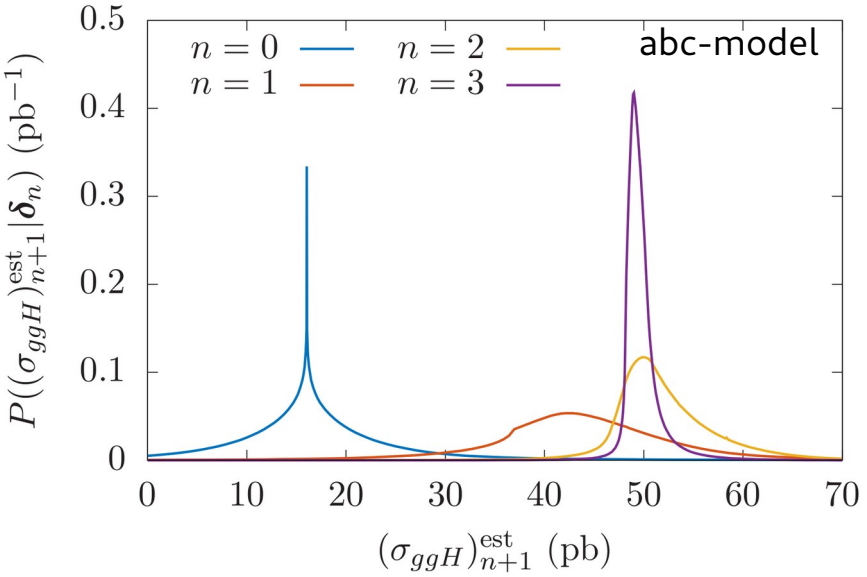
$$\mathcal{P}_{\rm sm}(\Sigma|\mathbf{\Sigma}_n) \approx \frac{\int d\mu\, P(\Sigma - \Sigma_n(\mu)|\mathbf{\Sigma}_n(\mu))\, P(\mathbf{\Sigma}_n(\mu))\, P_0(\mu)}{\int d\mu'\, P(\mathbf{\Sigma}_n(\mu'))\, P_0(\mu')}\,. \qquad \mu_{\rm FAC}$$

Scale average (the results are averaged with weight function)

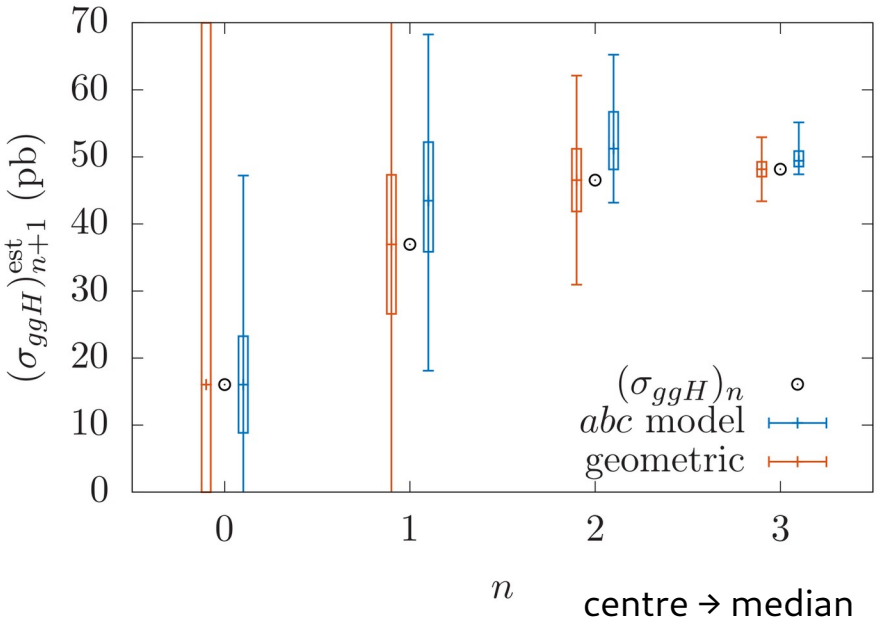
$$\mathcal{P}_{\rm sa}(\Sigma|\mathbf{\Sigma}_n) \approx \int d\mu\, w(\mu)\, P(\Sigma - \Sigma_n(\mu)|\mathbf{\Sigma}_n(\mu)) \qquad \mu_{\rm PMS}$$

Bayesian approach III

Example: Higgs production in gluon - fusion



Comparison of different unc. estimates:



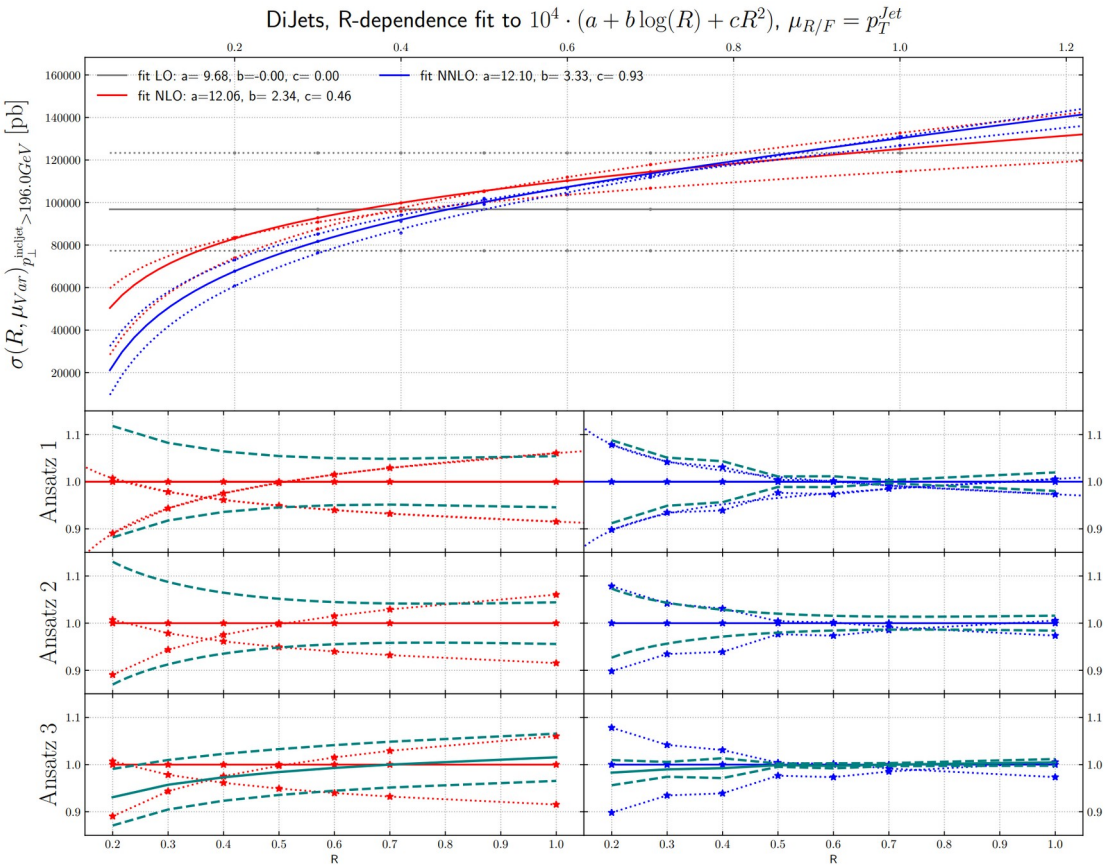
centre → median
error box → 68% CI
error bar → 95% CI

Example: inclusive jet production

→ Important process for PDF fits: sensitivity to gluon PDF at large-x

→ NNLO QCD corrections imply very small theory uncertainty

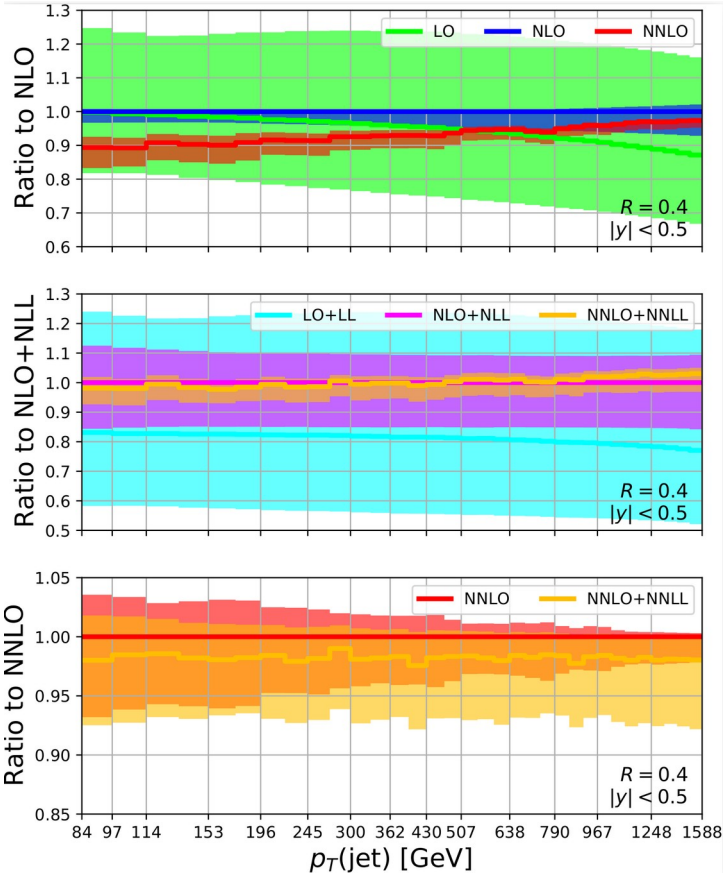
→ Significant jet radius dependence of uncertainties from scale variations



[1903.12563 Bellm et al]

Inclusive jet production: small-R resummation NNLO+NNLL

[Generet, Lee, Moult, Poncelet, Zhang'25]



FO scale variations

R=0.4

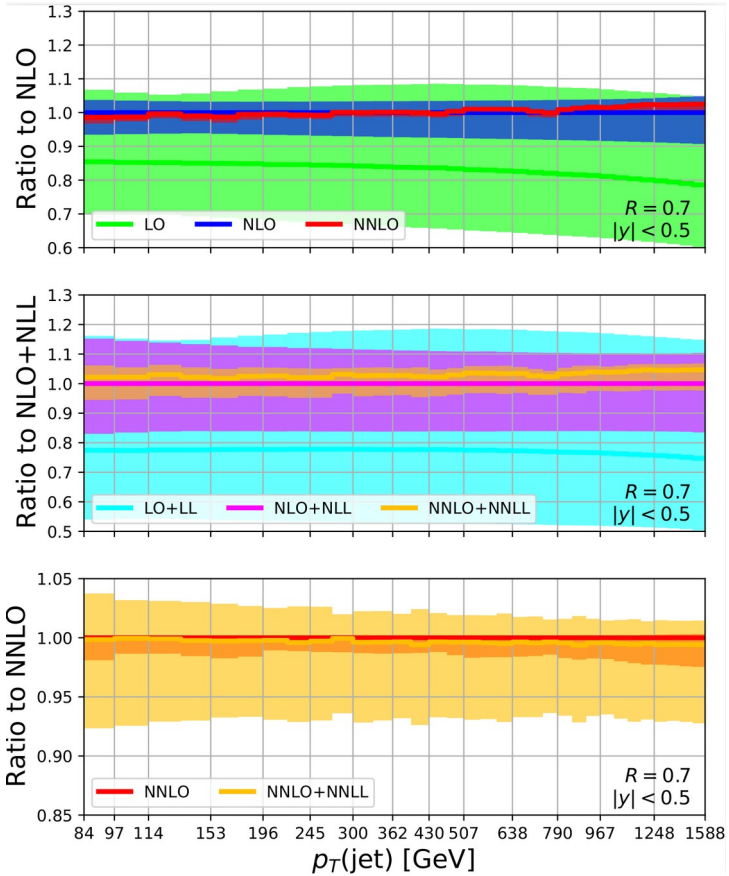
→ underestimation of NNLO correction

R=0.7

→ very small NNLO uncertainty

Resummation

→ stabilization of pert. series and uncertainties.

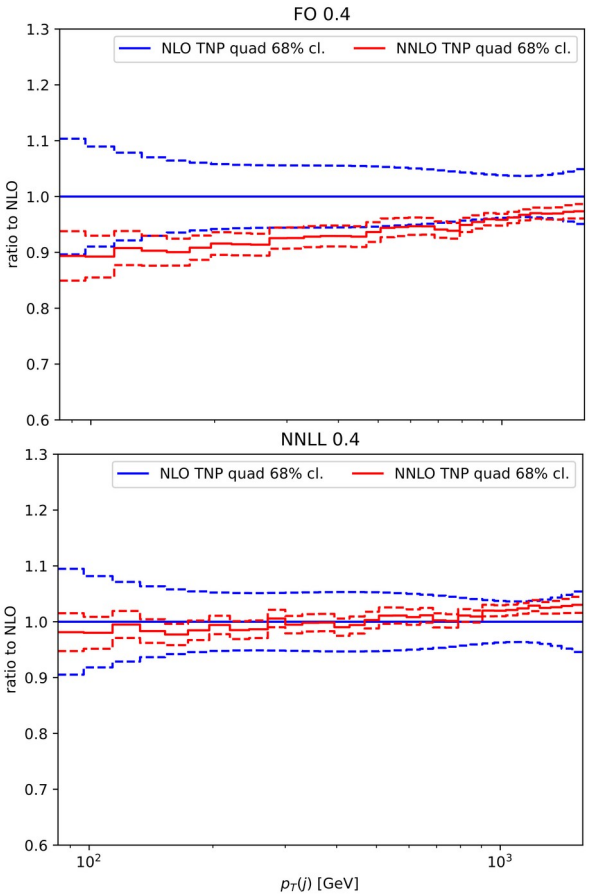


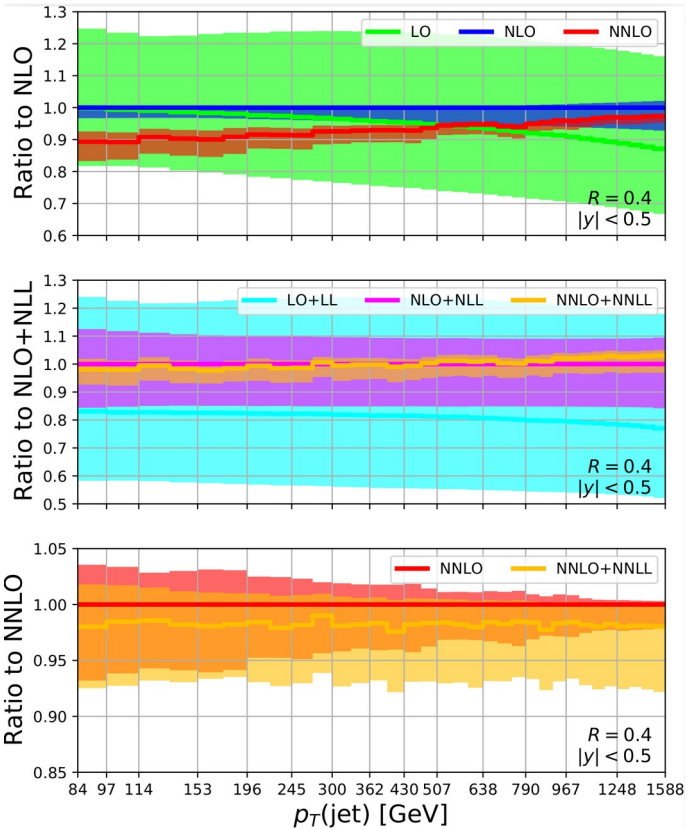
TNP uncertainties for inclusive jet production

R = 0.4

TNP uncertainties

- More sensible NLO uncertainties
- Similar to resummed scale variation



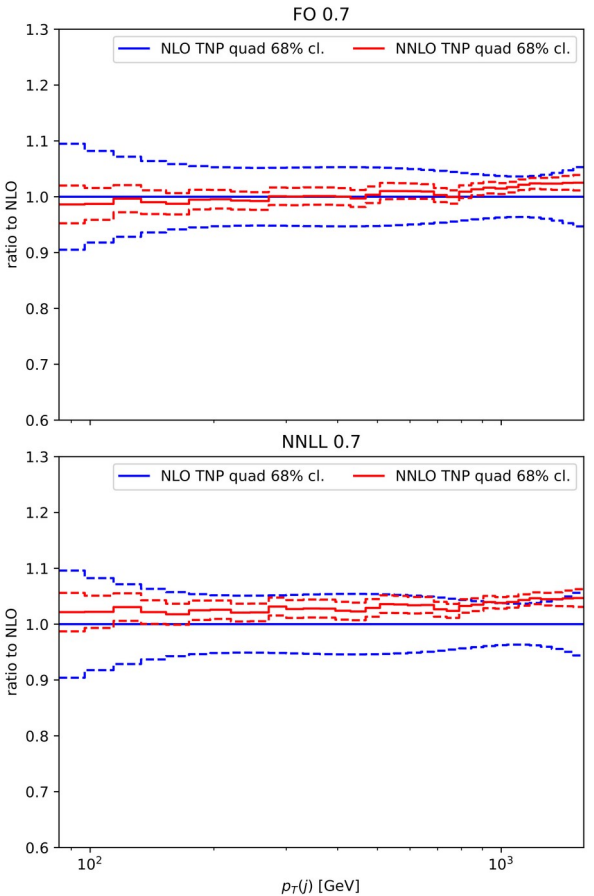


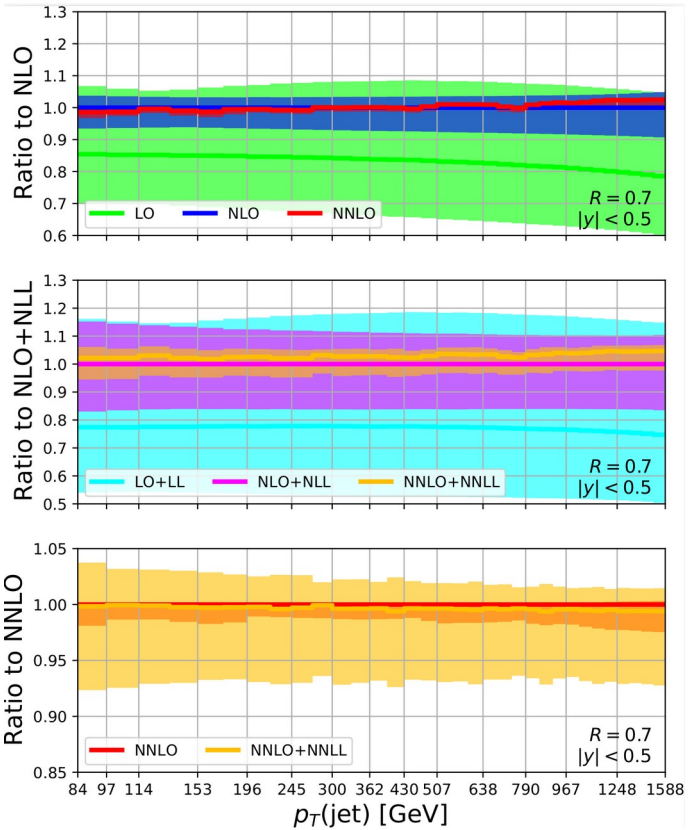
TNP uncertainties for inclusive jet production

R = 0.7

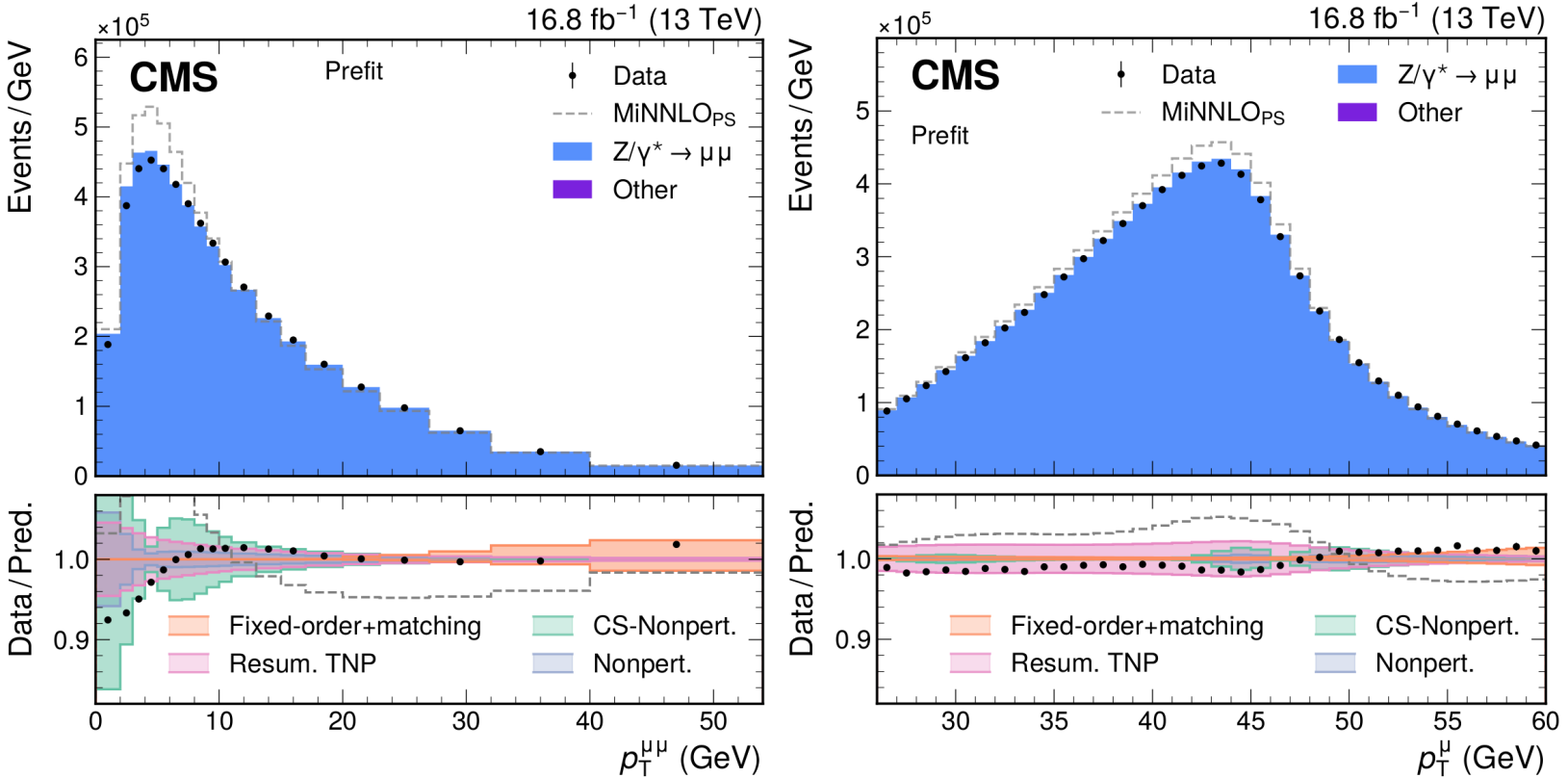
TNP uncertainties

- More sensible NNLO QCD uncertainties
- Similar to resummed scale variation

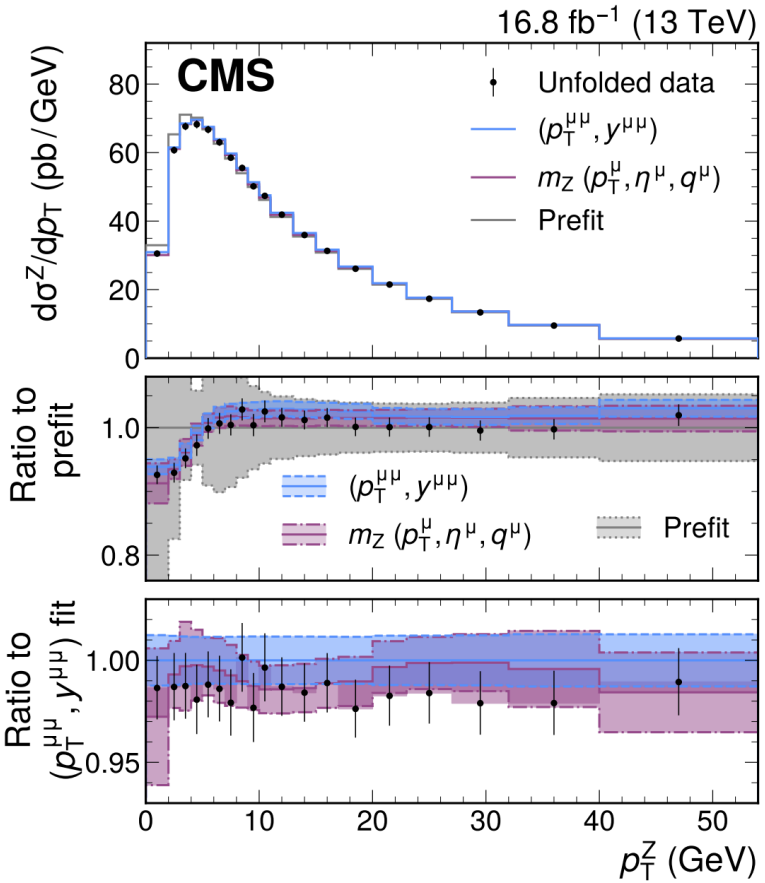




W-mass extraction: using Z as a test



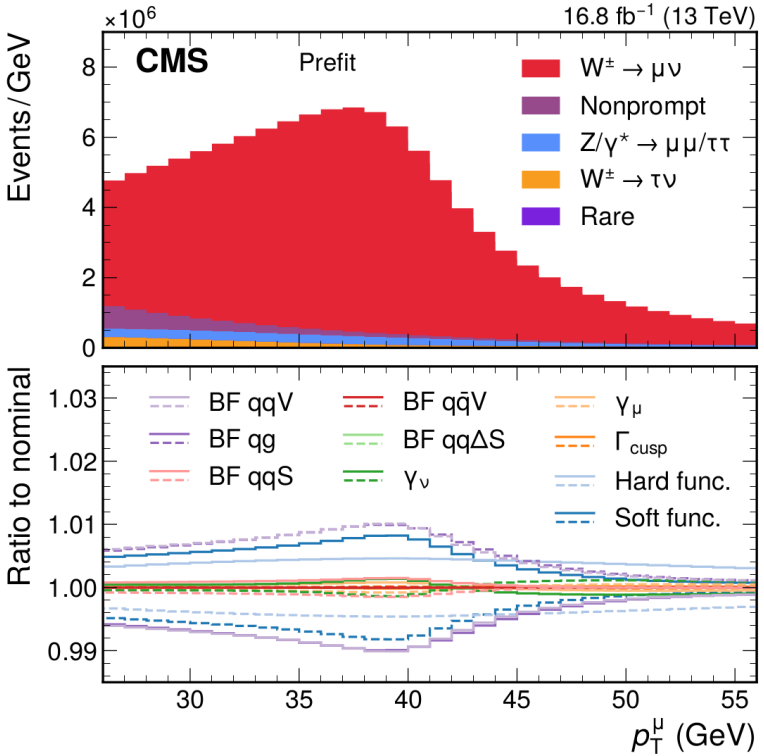
W-mass extraction: using Z as a test

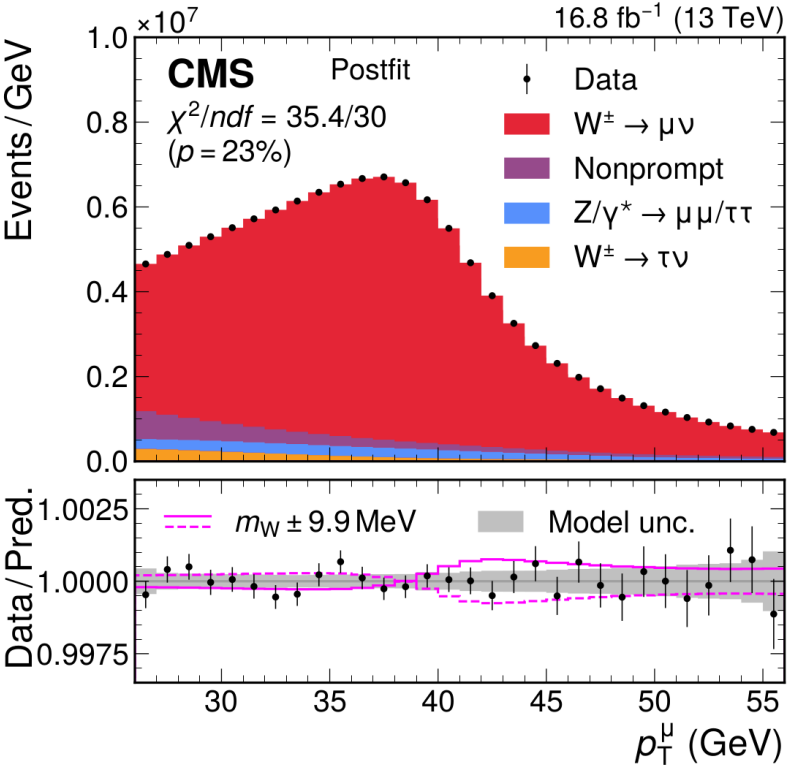


Using pT(Z) vs. W-like mZ fit
→ good consistency
→ demonstrates that pT(l)xeta(l)xq is sensitive enough for a competitive constraint vs pT(V)xy(V)
→ mZ consistent with PDG → internal consistency check

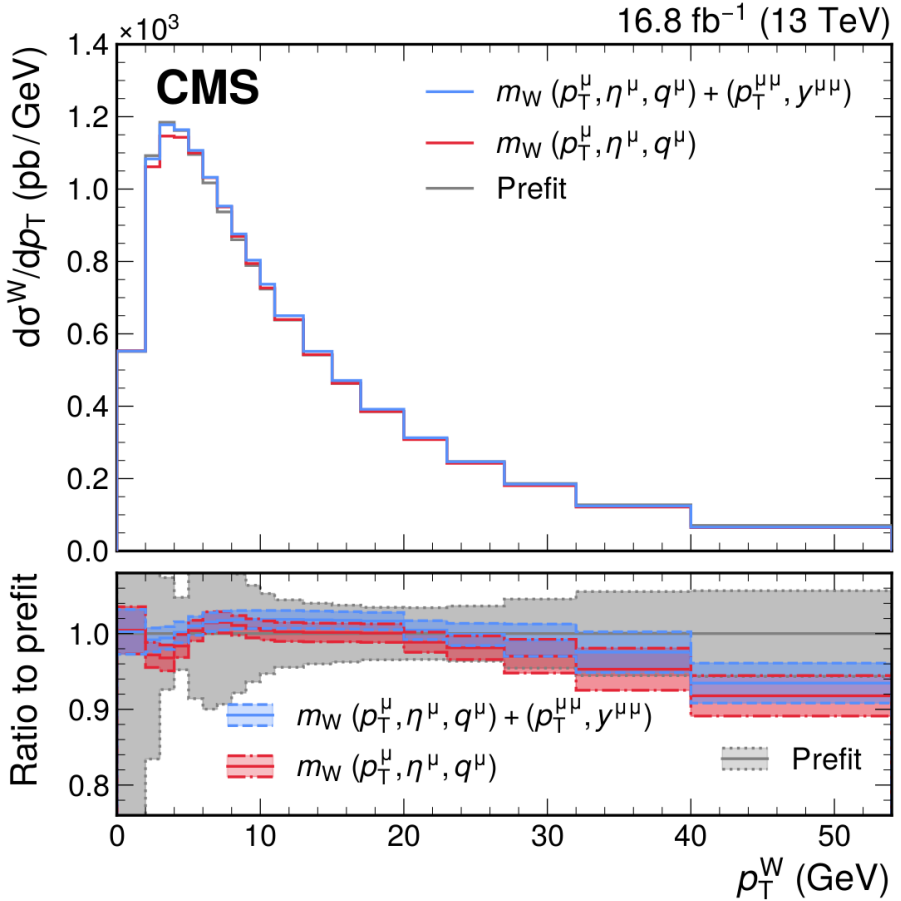
→ checked also different TNP schemes, i.e. ways to parameterize the higher order terms

W result


×10^6
16.8 fb^-1 (13 TeV)
CMS
Prefit
W± → μν
Nonprompt
Z/γ* → μμ/ττ
W± → τν
Rare
Events / GeV
Ratio to nominal
BF qqV
BF qg
BF qqS
BF qq̄V
BF qqΔS
γν
γμ
Γcusp
Hard func.
Soft func.
p_T^μ (GeV)


×10^7
16.8 fb^-1 (13 TeV)
CMS
Postfit
χ²/ndf = 35.4/30
(p = 23%)
Data
W± → μν
Nonprompt
Z/γ* → μμ/ττ
W± → τν
Events / GeV
Data / Pred.
mW ± 9.9 MeV
Model unc.
p_T^μ (GeV)

Combining W+Z?



Adding Z data:

→ only mildly impacts results

→ results are consistent: hints at consistent QCD modelling between W and Z

→ consistent constraints on non-pert. par.

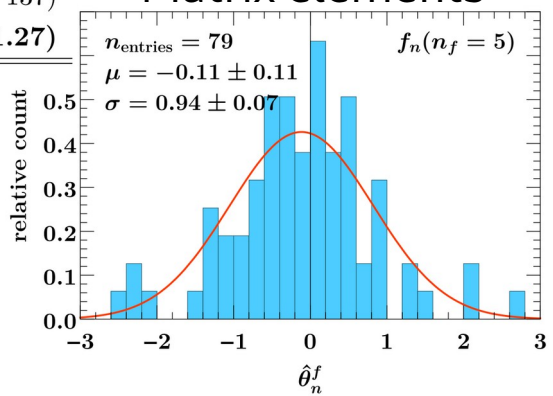
Priors for TNP parameters: example qT-resummation

[Tackman 2411.18606]

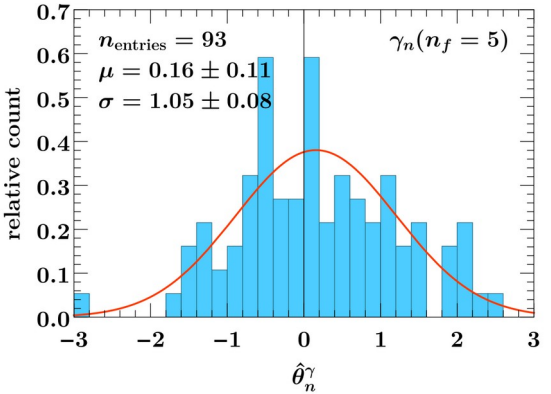
$\gamma(\alpha_s)$	N_n	$\hat{\gamma}_0/N_0$	$\hat{\gamma}_1/N_1$	$\hat{\gamma}_2/N_2$	$\hat{\gamma}_3/N_3$	$\hat{\gamma}_4/N_4$
β	1	-15.3	-77.3	-362	-9652	-30941
	4^{n+1}	-3.83	-4.83	-5.65	-37.7	-30.2
	$4^{n+1}C_F C_A^n$	$\mathbf{-1.28}$	$\mathbf{-0.54}$	$\mathbf{-0.21}$	$\mathbf{-0.47}$	$\mathbf{-0.12}$
γ_m	1	-8.00	-112	-950	-5650	-85648
	4^{n+1}	-2.00	-7.028	-14.8	-22.1	-83.6
	$4^{n+1}C_F C_A^n$	$\mathbf{-1.50}$	$\mathbf{-1.76}$	$\mathbf{-1.24}$	$\mathbf{-0.61}$	$\mathbf{-0.77}$
$2\Gamma^q_{\rm cusp}$	1	$+10.7$	$+73.7$	$+478$	$+282$	$(+140000)$
	4^{n+1}	$+2.67$	$+4.61$	$+7.48$	$+1.10$	$(+137)$
	$4^{n+1}C_F C_A^n$	$\mathbf{+2.00}$	$\mathbf{+1.15}$	$\mathbf{+0.62}$	$\mathbf{+0.03}$	$\mathbf{(+1.27)}$

"Statistic of many computations/loop orders"

Matrix elements

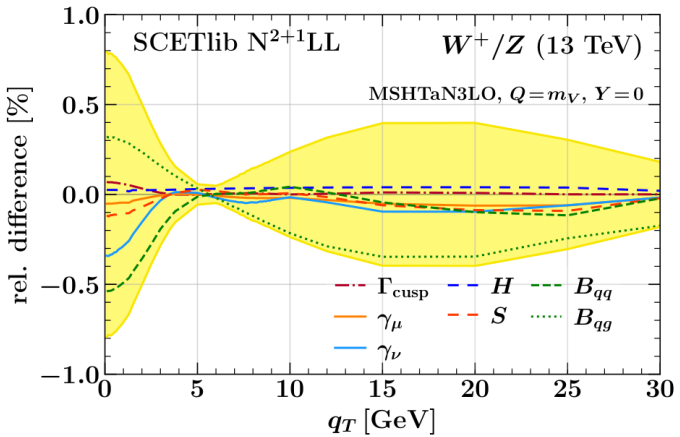


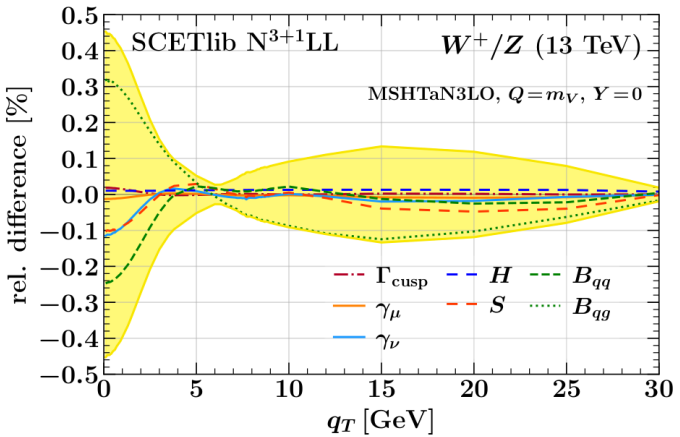
Anomalous dimensions



Turn this observation around to construct a **prior** for the **unknown** parameters
→ use variation as a priori estimates of MHOU

W/Z ratio





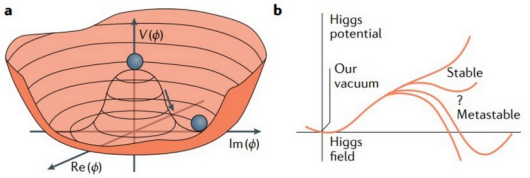
Many TNP variations are strongly correlated
→ cancellation in the ratio (similar to what on sees with scale variations)

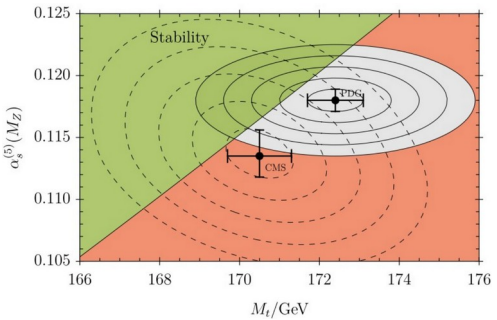
Caveat: this holds for the perturbative bits, non-perturbative effects might be different

Why ttbar + jet?

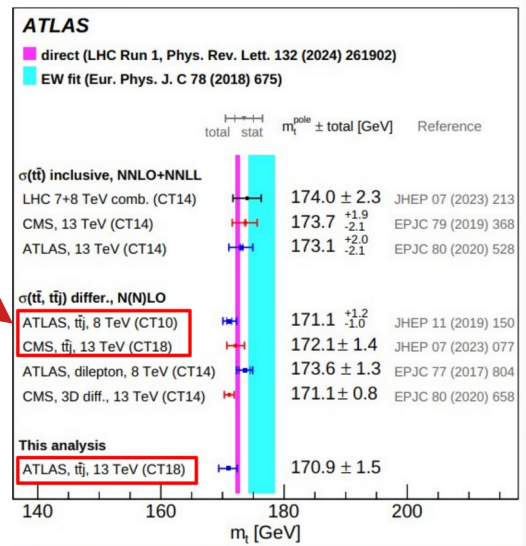
- 40% LHC top-quark pairs are produced in association with jets
  → many ttbar differential distributions are **directly sensitive** to ttbar+jet
- Additional opportunity to measure the **top-quark mass**, e.g. rho observable
  → precision determinations are crucial for SM tests
- QCD final state → important background for various rare SM processes or BSM searches

[Bass,DeRoeck,Kado Nat Rev Phys 3,608–624(2021)]





[Hiller,Hohne,Litim,Steudtner 2401.08811]



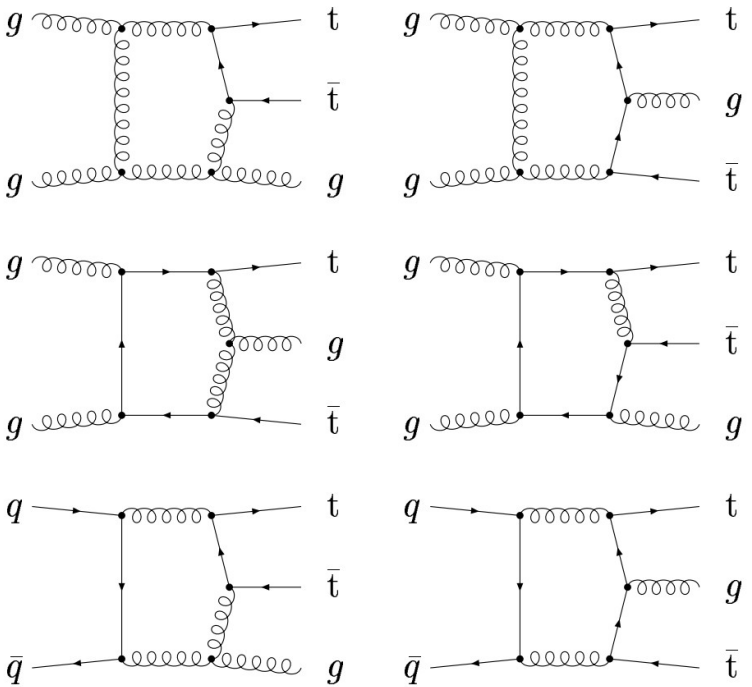
[ATLAS 2507.02632]

Theory predictions for top-quark pair + jet production I

NLO QCD (stable top)
- [Dittmaier,Uwer,Weinzierl (2007,2008)]

Merged NLO QCD+EW (stable top)
- [Gutschow,Lindert,Schonherr (2018)]

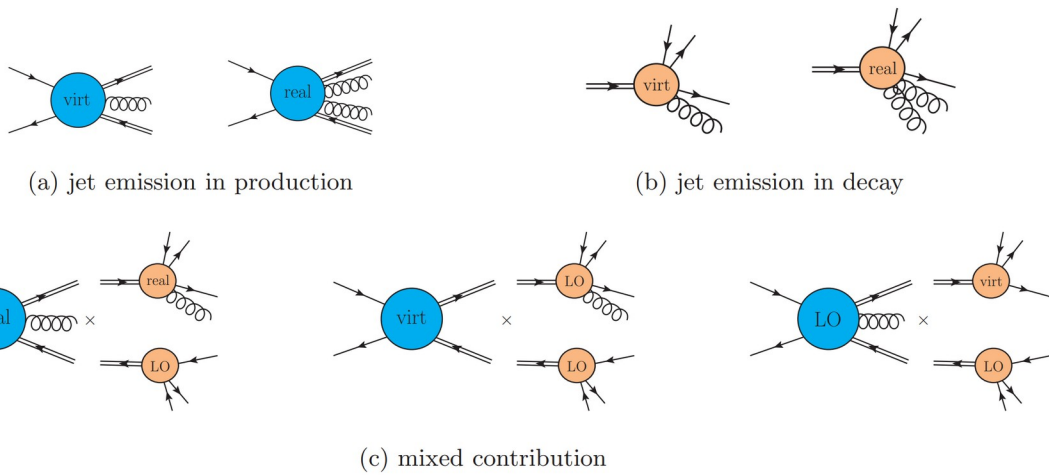
NLO QCD + PS
(+shower decays)
- [Kardos,Papadopoulos,Trocsanyi (2011)]
- [Alioli,Moch,Uwer (2011)]
- [Czakon,Hartanto,Kraus,Worek (2015)]



Theory predictions for top-quark pair + jet production II

NLO QCD (NWA)

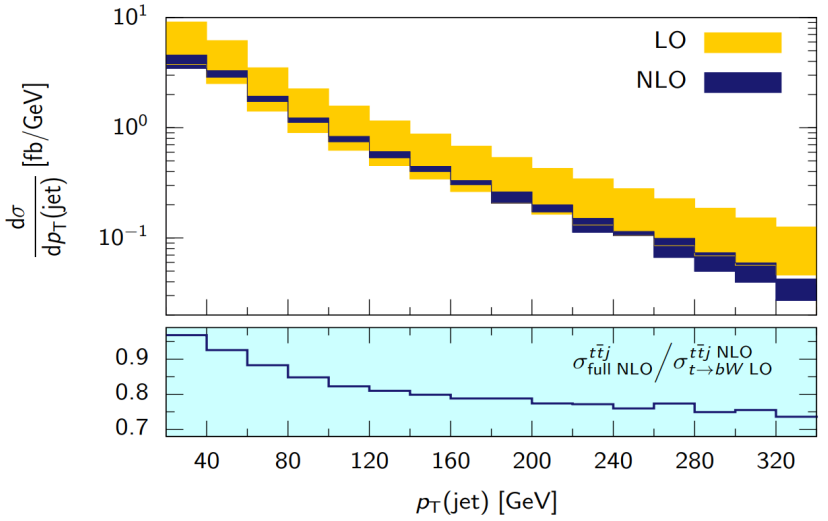
- [Melnikov,(Scharf),Schulze (2010,2011)]



(a) jet emission in production

(b) jet emission in decay

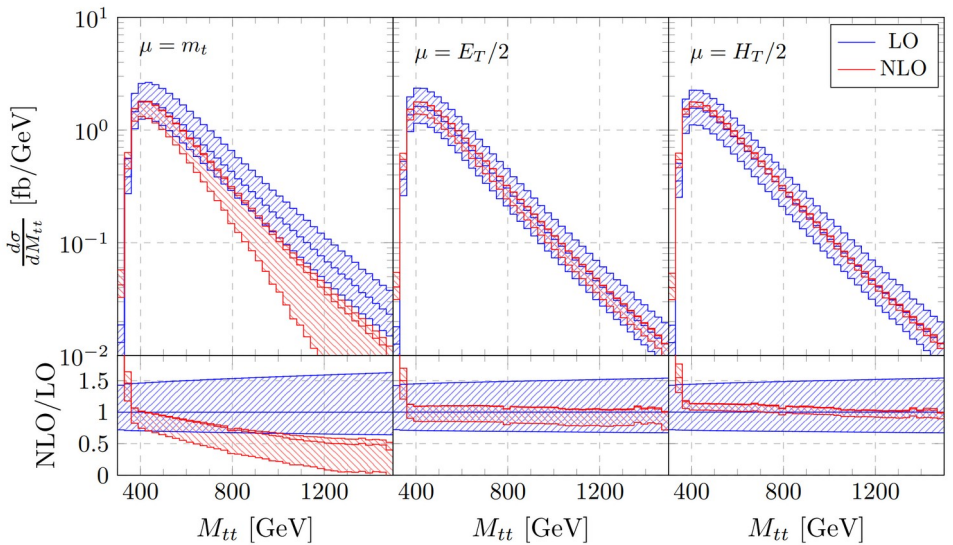
(c) mixed contribution



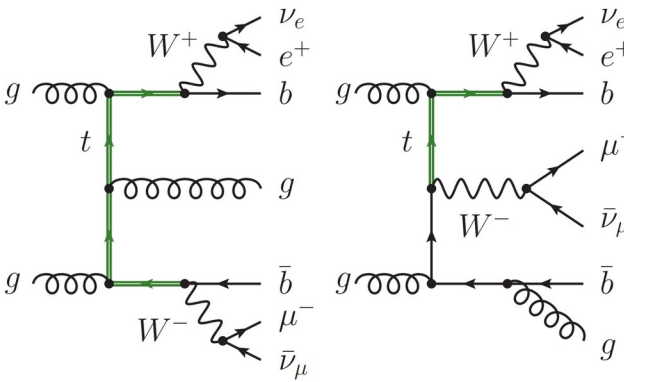
Theory predictions for top-quark pair + jet production III

NLO QCD (off-shell)

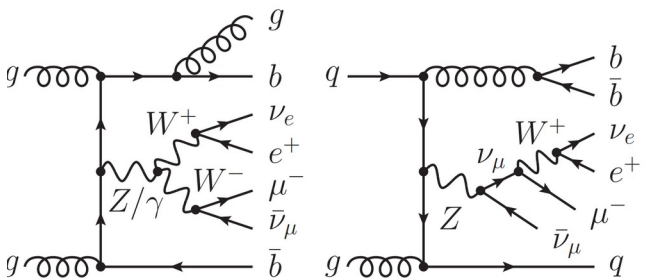
- [Bevilacqua,Hartanto,Kraus,Worek (2015,2016)]



double/single resonant



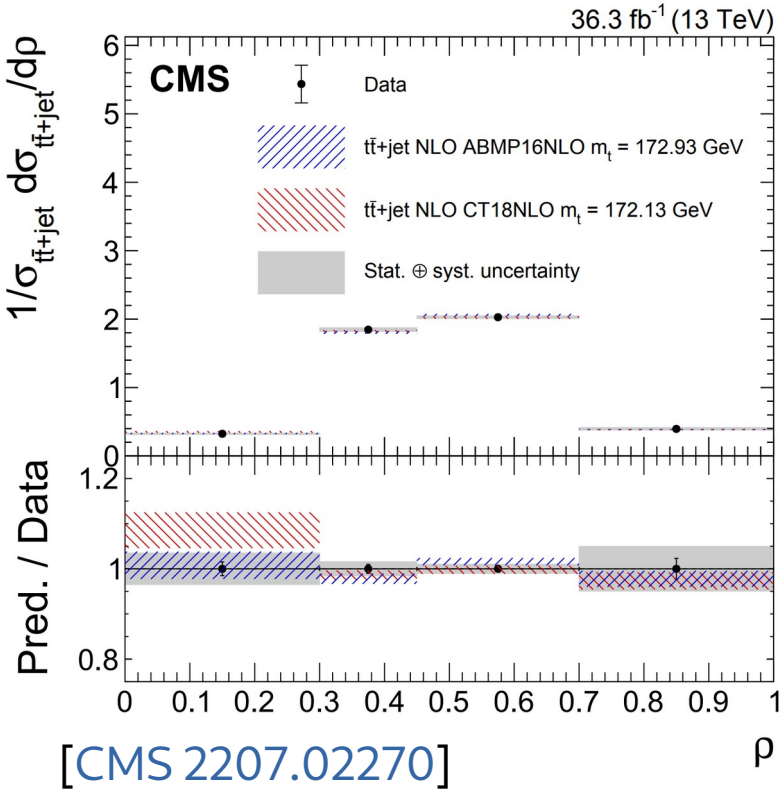
non resonant



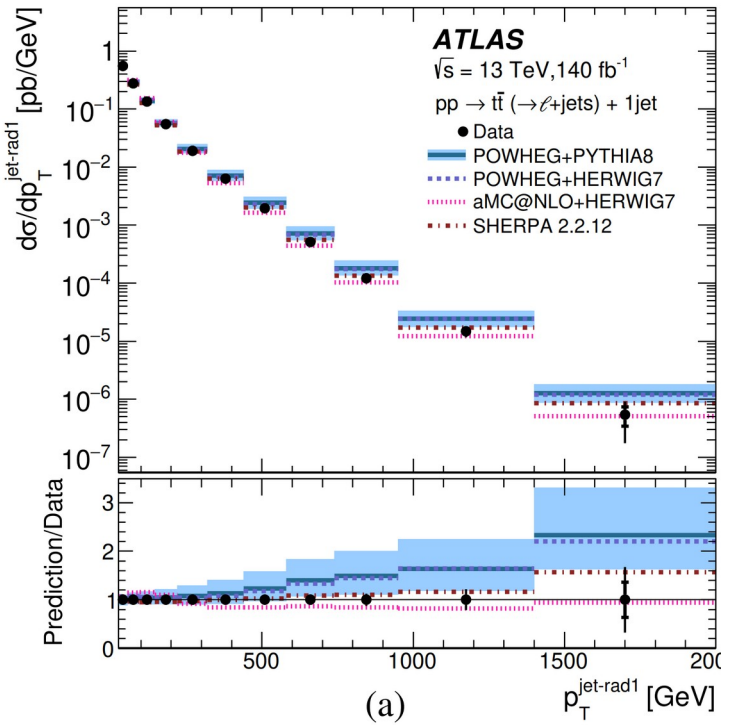
Why we need to go beyond NLO QCD?

Simple answer: the LHC experiments are doing an amazing job!

NLO QCD scale dependence sizeable compared to data uncertainties



[CMS 2207.02270]



[ATLAS 2406.19701]

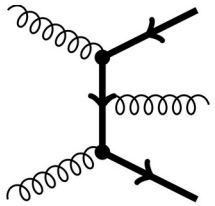
Perturbative QCD through NNLO QCD

$$\sigma_{h_1 h_2 \to X} = \sum_{ij} \int_0^1 \int_0^1 \mathrm{d}x_1 \mathrm{d}x_2 \phi_{i,h_1}(x_1, \mu_F^2) \phi_{j,h_2}(x_2, \mu_F^2) \hat{\sigma}_{ij \to X}(\alpha_s(\mu_R^2), \mu_R^2, \mu_F^2)$$

LO

α_s^3

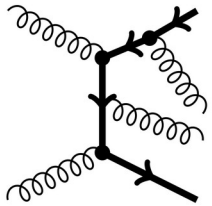
$\hat{\sigma}^{\mathrm{B}}$:



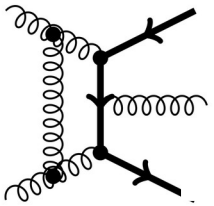
NLO

α_s^4

$\hat{\sigma}^{\mathrm{R}}$:



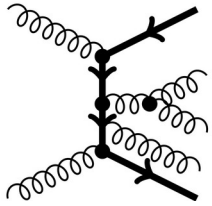
$\hat{\sigma}^{\mathrm{V}}$:



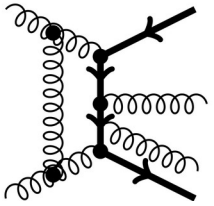
NNLO

α_s^5

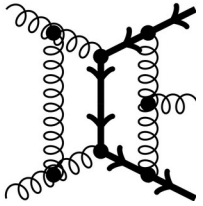
$\hat{\sigma}^{\mathrm{RR}}$:



$\hat{\sigma}^{\mathrm{RV}}$:



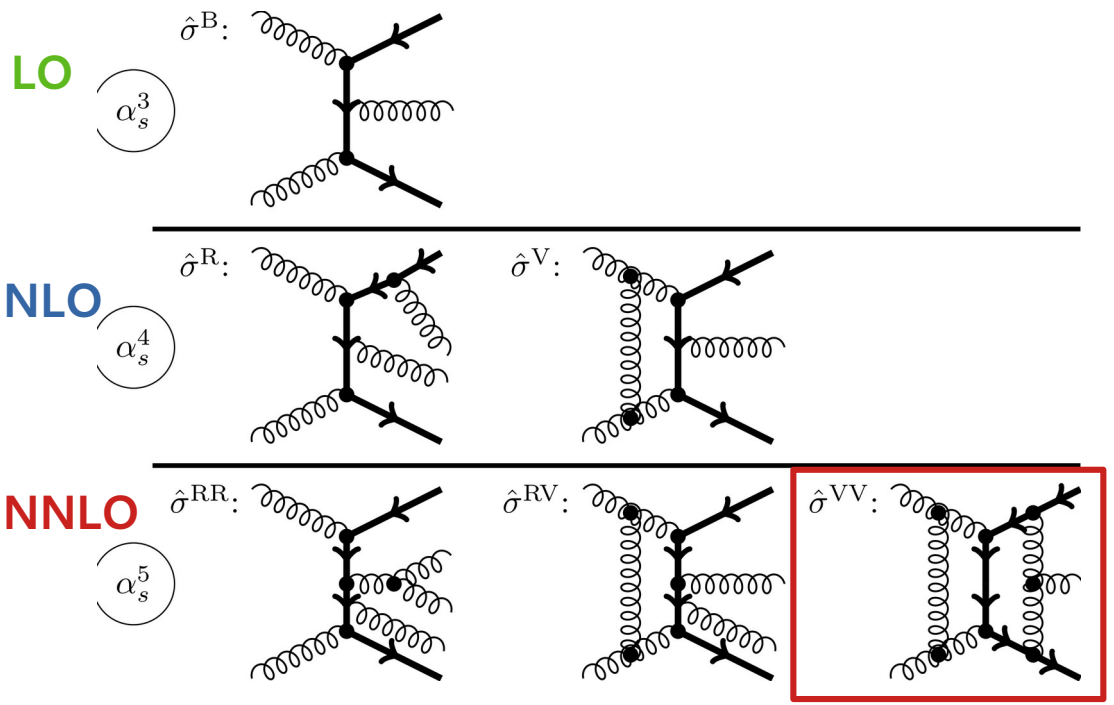
$\hat{\sigma}^{\mathrm{VV}}$:



Partonic cross section:

$$\hat{\sigma}_{ab \to X} = \hat{\sigma}^{(0)}_{ab \to X} + \hat{\sigma}^{(1)}_{ab \to X} + \hat{\sigma}^{(2)}_{ab \to X} + \mathcal{O}(\alpha_s^3)$$

Perturbative QCD through NNLO QCD



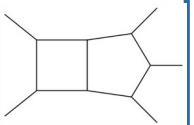
Multi-scale two-loop QCD amplitudes

→ **fast growing complexity:**
rational coef. and **special functions**

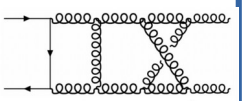
→ deeper understanding of the analytical properties

→ refinement of computational tools

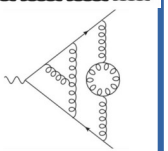
Two-loop 5-point
[Abreu, Agarwal, Badger, Buccioni, Chawdhry, Chicherin,Czakon, de Laurenties, Febres-Cordero, Gambuti, Gehrmann, Henn, Lo Presti, Manteuffel, Ma, Mitov, Page, Peraro, Poncelet, Schabiner Sotnikov, Tancredi, Zhang,...]



Three-loop 4-point
[Bargiela, Dobadilla, Canko, Caola, Jakubcik, Gambuti, Gehrmann, Henn, Lim, Mella, Mistleberger, Wasser, Manteuffel, Syrrakow, Smirnov, Trancredi, ...]



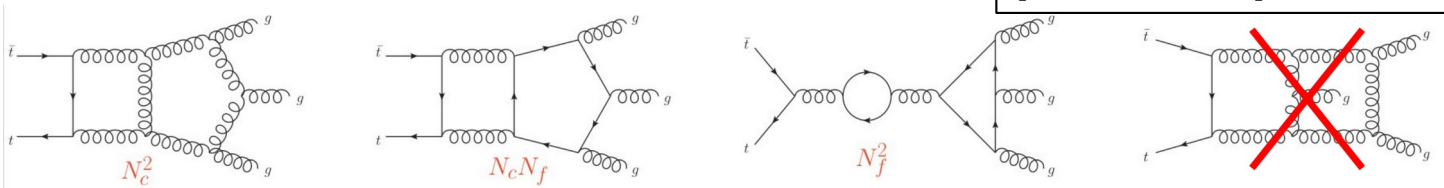
Four-loop 3-point
[Henn, Lee, Manteuffel, Schabinger, Smirnov, Steinhauser,...]



Virtual amplitudes

Double virtual QCD corrections to $t\bar{t}$+jet production at the LHC, Badger, Becchetti, Brancaccio, Czakon, Hartanto, **Poncelet**, Zoia [arxiv:2511.11424]

Sample diagrams



Colour & spin structures

$$\mathcal{M}^{(L)}\left(1_{\bar{t}},2_t,3_g,4_g,5_g\right) = \sqrt{2}\,\bar{g}_s^3\left[(4\pi)^\epsilon e^{-\epsilon\gamma_E}\frac{\bar{\alpha}_s}{4\pi}\right]^L \times \sum_{\sigma\in Z_3}\left(t^{a_{\sigma(3)}}t^{a_{\sigma(4)}}t^{a_{\sigma(5)}}\right)_{i_2}^{\bar{i}_1}\,\mathcal{A}_g^{(L)}\left(1_{\bar{t}},2_t,\sigma(3)_g,\sigma(4)_g,\sigma(5)_g\right)$$

Retaining full spin information
→ allows to incorporate spin-correlations in NWA decays

$$\mathcal{A}_x^{(L)h_3h_4h_5} = \sum_{i=1}^{4}\Psi_i\,\mathcal{G}_{x;i}^{(L)h_3h_4h_5}$$

Leading colour expansion of partial amplitudes:

$$\mathcal{F}_x^{(1)} = N_c\,F_x^{(1),N_c} + n_f\,F_x^{(1),n_f}\,,$$
$$\mathcal{F}_x^{(2)} = N_c^2\,F_x^{(2),N_c^2} + N_c n_f\,F_x^{(2),N_c n_f} + n_f^2\,F_x^{(2),n_f^2}\,.$$

Partial amplitudes: masters expressed in basis functions

$$\tilde{\mathcal{F}}_{x,i}^{(L)h_1h_2h_3} = \sum_j d_j(\{p\},\epsilon)\mathrm{MI}_j(\{p\},\epsilon)$$

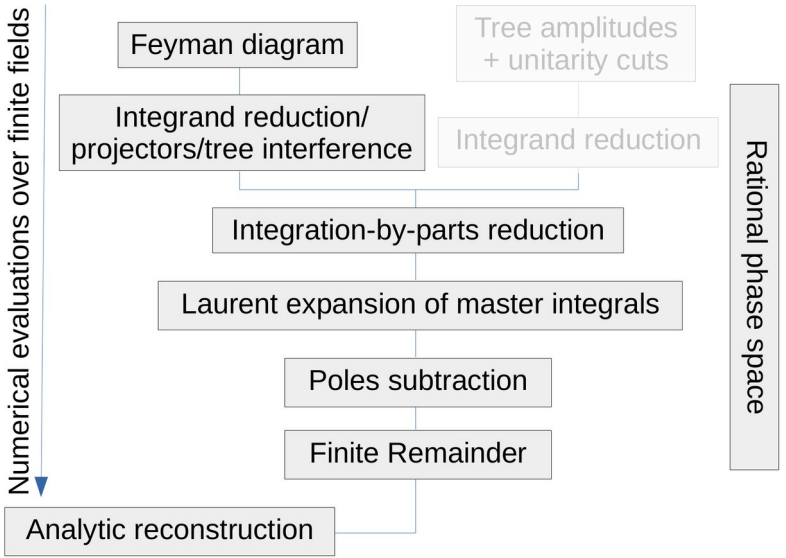
$$\tilde{\mathcal{F}}_{x;i}^{(L)h_3h_4h_5} = \sum_k r_{x;i,k}^{(L)h_3h_4h_5}\,m_k(\vec{f})$$

algebraic

rational

Reconstruction of rational parts

Workflow



QGRAF[Nogueira], FORM[Vermaseren,etal] MATHEMATICA, SPINNEY[Cullen,etal]

finite field framework: FINITEFLOW[Peraro(2019)]

IBP identities generated using LITERED[Lee(2012)] solved numerically in FINITEFLOW using Laporta algorithm[Laporta(2000)]

Credit: Bayu

Mature technology + new optimizations

- Syzygy and module intersection techniques to simplify IBPs [NeatIBP]
- Exploitation of Q-linear relations
- Denominator Ansaetze
- on-the-fly univariate partial fraction

	1	2	3	4
master integral coefficients of mass-renormalised amplitude (full ϵ dependence)	404/393	398/389	411/402	421/411
special function coefficients of finite remainder	314/303	305/296	321/312	326/317
linear relations	291/280	287/278	299/293	304/299
denominator matching #1	291/0	287/0	299/0	304/0
partial fraction decomposition in x_{5123}	44/40	55/51	57/54	58/54
denominator matching #2	44/0	54/0	54/0	56/0
number of sample points (1 prime field)	137076	89624	161482	179838

Algebraic part → special functions

Differential Equations: $\mathrm{d}\vec{\mathrm{MI}} = \mathrm{d}A(\{p\},\epsilon)\vec{\mathrm{MI}}$

Canonical basis: $\mathrm{d}\vec{\mathrm{MI}} = \epsilon\mathrm{d}\tilde{A}(\{p\})\vec{\mathrm{MI}}$

[Remiddi, 97]
[Gehrmann, Remiddi, 99]
[Henn, 13]

→ not available in ttj case (elliptic integrals):
However, still translates into a system of DEQs for basis functions: $\mathrm{d}G(\vec{d}) = M(\vec{d}) \cdot G(\vec{d})$

$$G(\vec{d}) = \begin{pmatrix} f_i^{(4^*)} \\ f_i^{(4)} \\ \text{weight-3} \\ \text{weight-2} \\ f_i^{(1)} \\ 1 \end{pmatrix}, \qquad M(\vec{d}) = \begin{pmatrix} Y_{4^*,4^*} & 0 & Y_{4^*,3} & Y_{4^*,2} & 0 & 0 \\ 0 & 0 & X_{4,3} & 0 & 0 & 0 \\ 0 & 0 & 0 & X_{3,2} & 0 & 0 \\ 0 & 0 & 0 & 0 & X_{2,1} & 0 \\ 0 & 0 & 0 & 0 & 0 & X_{1,0} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Solve numerically from pre-computed boundary points [AMFlow]

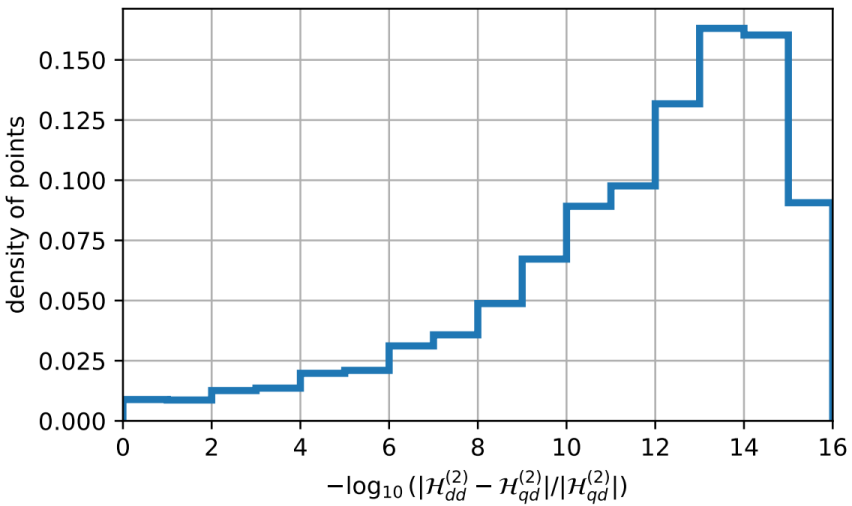
Numerical evaluation of master integrals

→ Solve DEQ by direct integration
(Bulirsch-Stoer algorithm)
→ Error control based on each function's contribution to squared matrix element

$$\mathcal{Q} \equiv \frac{1}{(4\pi)^4} \frac{2 \sum_{\text{colour}} \sum_{\text{pol.}} \mathcal{R}^{(0)*} \mathcal{R}^{(2)}}{\sum_{\text{colour}} \sum_{\text{pol.}} \left|\mathcal{R}^{(0)}\right|^2}$$

$$\Delta\mathcal{Q}(t) = \left[\sum_{i=1}^{947} \left| \frac{\partial Q}{\partial f_i} \right|^2_{\vec{f}=\vec{f}(t)} \left|\Delta f_i(t)\right|^2 \right]^{1/2}$$

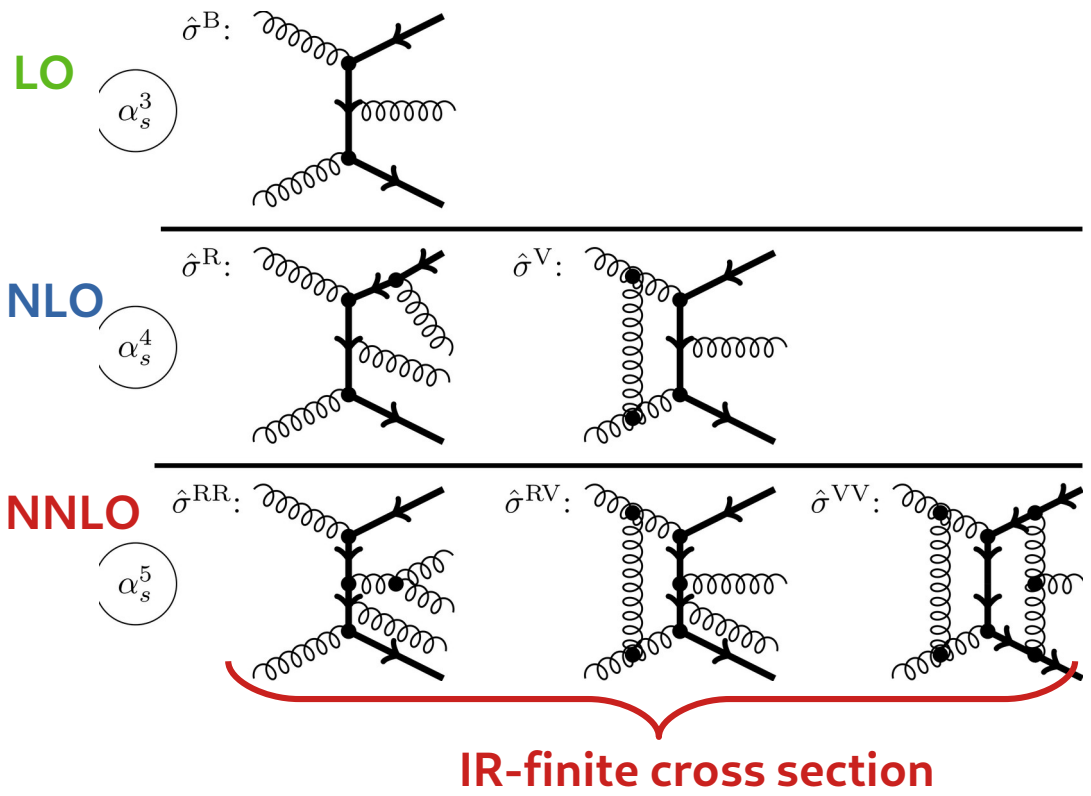
Channel	Functions [s]	Coefficents [s]	Assembly [s]	total [s]
$gg \to \bar{t}tg$	2.69	30.90	1.58	35.17
$\bar{q}g \to \bar{t}t\bar{q}$	2.16	9.40	0.18	11.74
$qg \to \bar{t}tq$	2.50	9.62	0.21	12.33
$q\bar{q} \to \bar{t}tg$	2.12	9.30	0.18	11.60



→ directly control numerical error of the final result
→ ready for phenomenology

C++ code publicly available

Perturbative QCD through NNLO QCD



Infrared finite differential cross sections at NNLO QCD:

~20 years to solve this problem

→ highly non-trivial IR structure

→ plethora of subtraction schemes

qT-slicing [Catani'07]
N-jettiness slicing [Gaunt'15/Boughezal'15]
Antenna [Gehrmann'05-'08]
Colorful [DelDuca'05-'15]
Projetction [Cacciari'15]
Geometric [Herzog'18]
Unsubtraction [Aguilera-Verdugo'19]
Nested collinear [Caola'17]
Local Analytic [Magnea'18]
Sector-improved residue subtraction [Czakon'10-'14,'19]

Small interlude: sub-leading colour in 2 → 3 processes

How much color do we really need? Two-loop subleading-color effects in photon and jet physics [Czakon, Poncelet 2512.17591]

	Leading colour amplitudes	Full colour amplitudes
$pp \to \gamma\gamma\gamma$	[1911.00479][2010.15834] [2012.13553]	[2305.17056]
$pp \to \gamma\gamma j$	[2102.01820][2103.04319]	[2105.04585]
$pp \to \gamma j j$		[2304.06682]
$pp \to jjj$	[1904.00945] [2102.13609]	[2311.09870] [2311.10086] [2311.18752]

Approximation made in 2 → 3 calculations:

$$\mathcal{F}^{(2)}(\mu_R^2) = 2\,\mathrm{Re}\left[\mathcal{M}^{\dagger(0)}\mathcal{F}^{(2)}\right](\mu_R^2) + \left|\mathcal{F}^{(1)}\right|^2(\mu_R^2) \equiv \mathcal{F}^{(2)}(s_{12}) + \sum_{i=1}^{4} c_i \ln^i\left(\frac{\mu_R^2}{s_{12}}\right)$$

$$F^{(2)}_{\text{l.c. resc.}}(\mu^2) = \frac{F^{(0)}}{F^{(0)}_{\text{l.c.}}} F^{(2)}_{\text{l.c.}}(Q^2) + \sum_{i=0}^{4} c_i \log^i(\mu^2/Q^2)$$

Parametrise the sub-leading colour terms

For example $pp \to \gamma\gamma j$

$$\frac{2\,\mathrm{Re}\,\langle \mathcal{M}^{(0)}|\mathcal{F}^{(\overline{\mathrm{MS}},2)}(\mu=\hat{E}_{\mathrm{CMS}})\rangle}{\langle \mathcal{M}^{(0)}|\mathcal{M}^{(0)}\rangle}$$

$$= N_c^2\left[c_{1,0} + \frac{1}{N_c^2}c_{1,1} + \frac{1}{N_c^4}c_{1,2} + \frac{n_f}{N_c}\left(c_{2,0} + \frac{1}{N_c^2}c_{2,1}\right)\right.$$
$$\left. + \sum_{q'}\frac{Q_{q'}}{Q_q}\,N_c\left(c_{3,0} + \frac{1}{N_c^2}c_{3,1}\right) + \sum_{q'}\frac{Q_{q'}^2}{Q_q^2}\,N_c\left(c_{4,0} + \frac{1}{N_c^2}c_{4,1} + \frac{n_f}{N_c}\,c_{5,0}\right)\right)\Bigg]$$

full colour structure

$$\approx N_c^2\left[c_{1,0} + \frac{n_f}{N_c}c_{2,0}\right] + \sum_{q'}\frac{Q_{q'}^2}{Q_q^2}\,n_f\,c_{5,0}.$$

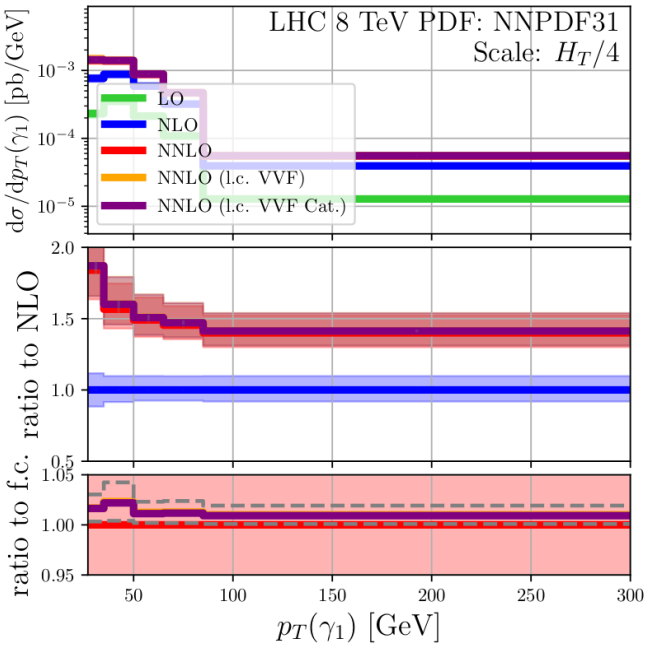
Leading-colour / planar approximation

Parametrise the leading missing term:

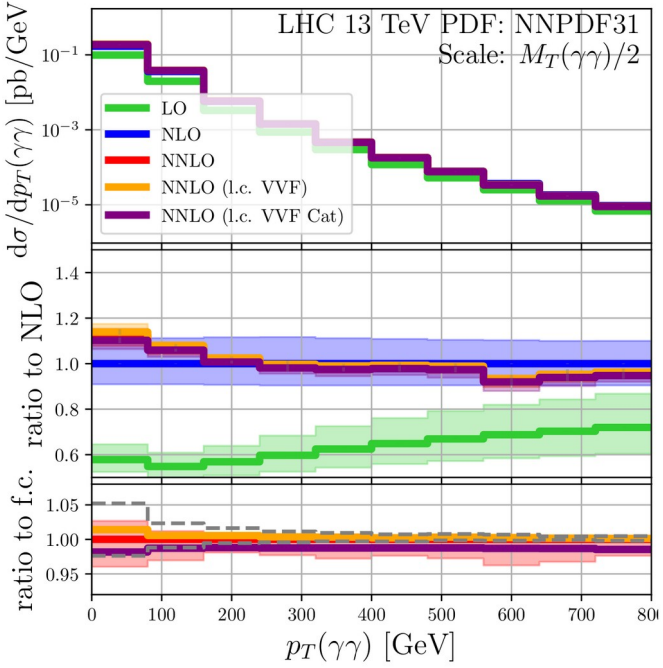
$$\left.\frac{2\,\mathrm{Re}\,\langle \mathcal{M}^{(0)}|\mathcal{F}^{(\overline{\mathrm{MS}},2)}(\mu=\hat{E}_{\mathrm{CMS}})\rangle}{\langle \mathcal{M}^{(0)}|\mathcal{M}^{(0)}\rangle}\right|^{\theta}_{\mathrm{l.c.}} = N_c^2\left[c_{1,0} + \frac{n_f}{N_c}c_{2,0}\right] + \sum_{q'}\frac{Q_{q'}^2}{Q_q^2}\,n_f\,c_{5,0} + \sum_{q'}\frac{Q_{q'}^2}{Q_q^2}N_c\theta\,c_{1,0} \qquad \theta \in [-1,1]$$

Sub-leading colour estimates vs. reality

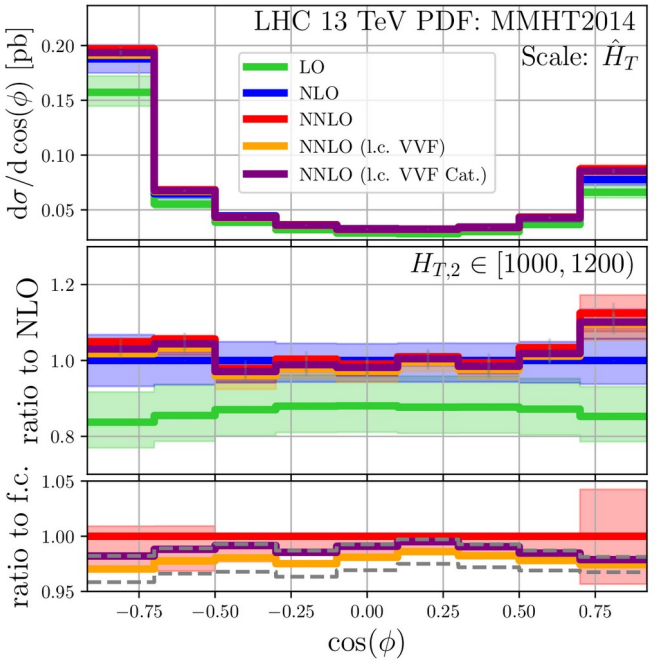
$pp \to \gamma\gamma\gamma$



$pp \to \gamma\gamma j$



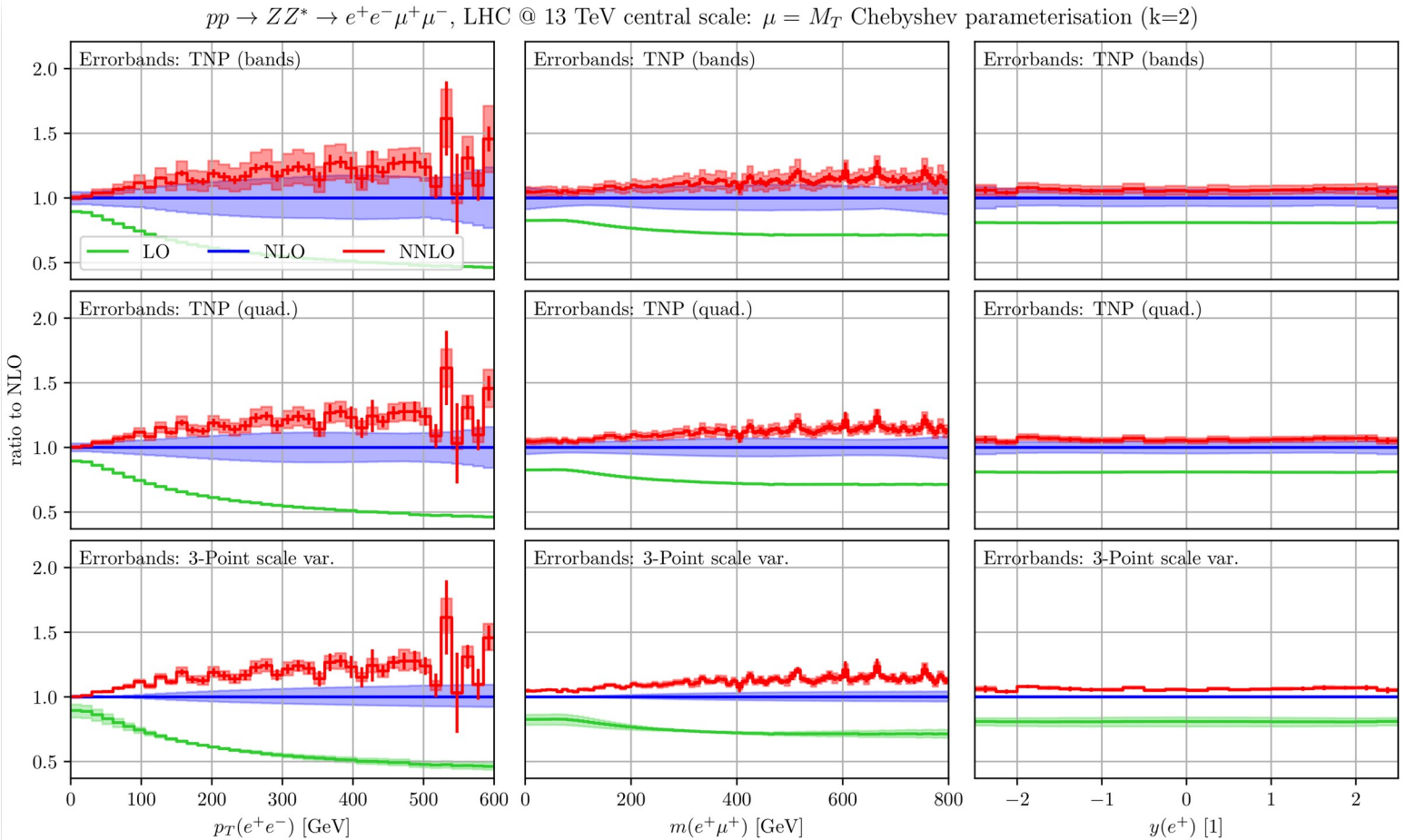
$pp \to jjj$



Grey dashed lines: envelopes of variation

→ considering the simplicity of the Ansatz **satisfying results**!

Uncertainties from TNPs - ZZ

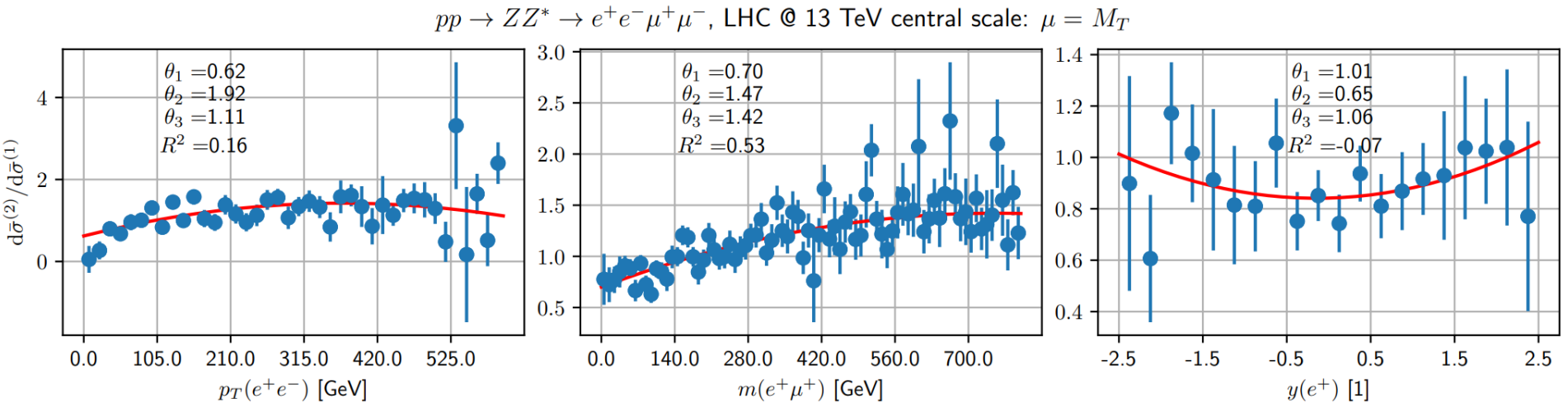


Band: sample
$\theta \in [-1, 1]$
Quad: add individual
$\theta = \pm 1$
in quadrature
(→ 65% CL)

Example of TNP fit: pp → ZZ

$$\frac{\mathrm{d}\bar{\sigma}^{(N+1)}_{\mathrm{TNP}}}{\mathrm{d}\bar{\sigma}^{(0)}} = \sum_{j=1}^{N} f_k^{(j)}\left(\vec{\theta}, x\right)\left(\frac{\mathrm{d}\bar{\sigma}^{(j)}}{\mathrm{d}\bar{\sigma}^{(0)}}\right)$$

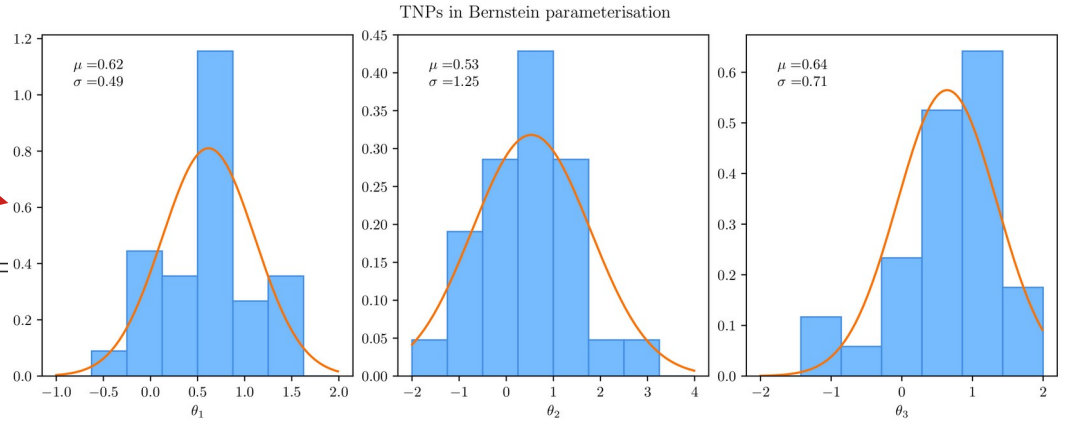
$$f_k^B\left(\vec{\theta}, x\right) = \sum_{i=0}^{k} \theta_i \binom{k}{i} x^{k-i}(1-x)^i$$



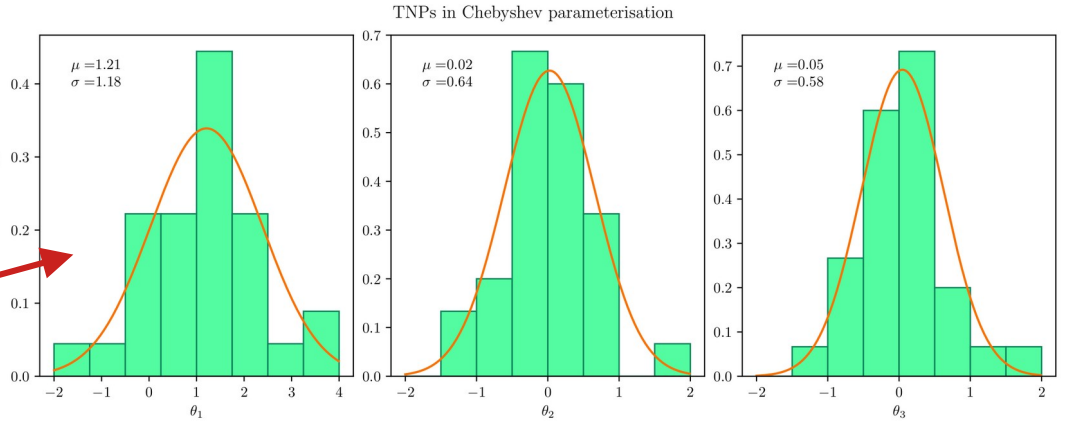
Fits - meta-study

$$f_k^B\left(\vec{\theta}, x\right) = \sum_{i=0}^{k} \theta_i \binom{k}{i} x^{k-i}(1-x)^i$$

Process	Distributions
$pp \to H$ (full theory)	y_H
$pp \to ZZ^* \to e^+e^-\mu^+\mu^-$	$M_{e^+\mu^+}$, $p_T^{e^+e^-}$, y_{e^+}
$pp \to WW^* \to e\nu_e\mu\nu_\mu$	M_{WW}, $p_T^{\mu^-}$, y_{W^-}
$pp \to (W \to \ell\nu) + c$	p_T^{ℓ} ,$\lvert y_\ell \rvert$,
$pp \to t\bar{t}$	$M_{t\bar{t}}$, p_T^t, y_t
$pp \to t\bar{t} \to b\bar{b}\ell\bar{\ell}$	$M_{\ell\bar{\ell}}$, $p_T^{b_1}$, $p_{T,\ell_1}/p_{T,\ell_2}$
$pp \to \gamma\gamma$	$M_{\gamma\gamma}$, $p_T^{\gamma_1}$, $y_{\gamma\gamma}$
$pp \to \gamma\gamma j$	$M_{\gamma\gamma}$, $p_T^{\gamma\gamma}$, $\cos\phi_{\rm CS}$, $\lvert y_{\gamma_1} \rvert$, $\Delta\phi_{\gamma\gamma}$
$pp \to jjj$	EEC with $H_{T,2} \in [1000, 1500), [1500, 2000), [3500, \infty)$
$pp \to \gamma jj$	$M_{\gamma jj}$, p_T^j, $\lvert y_{\gamma-\rm jet} \rvert$, $E_{T,\gamma}$

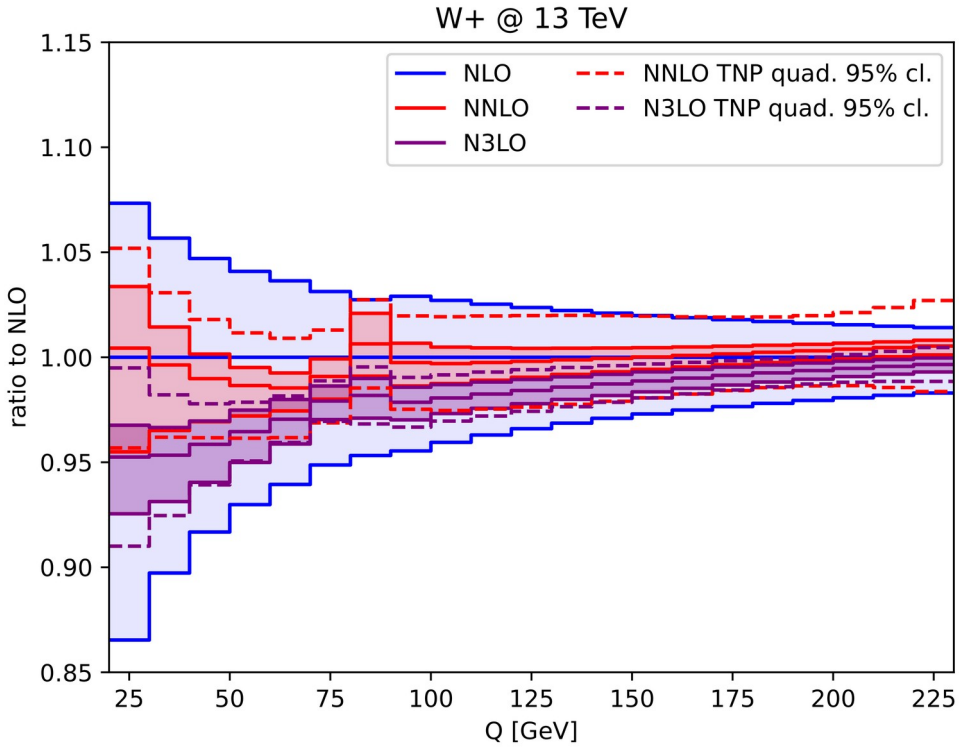


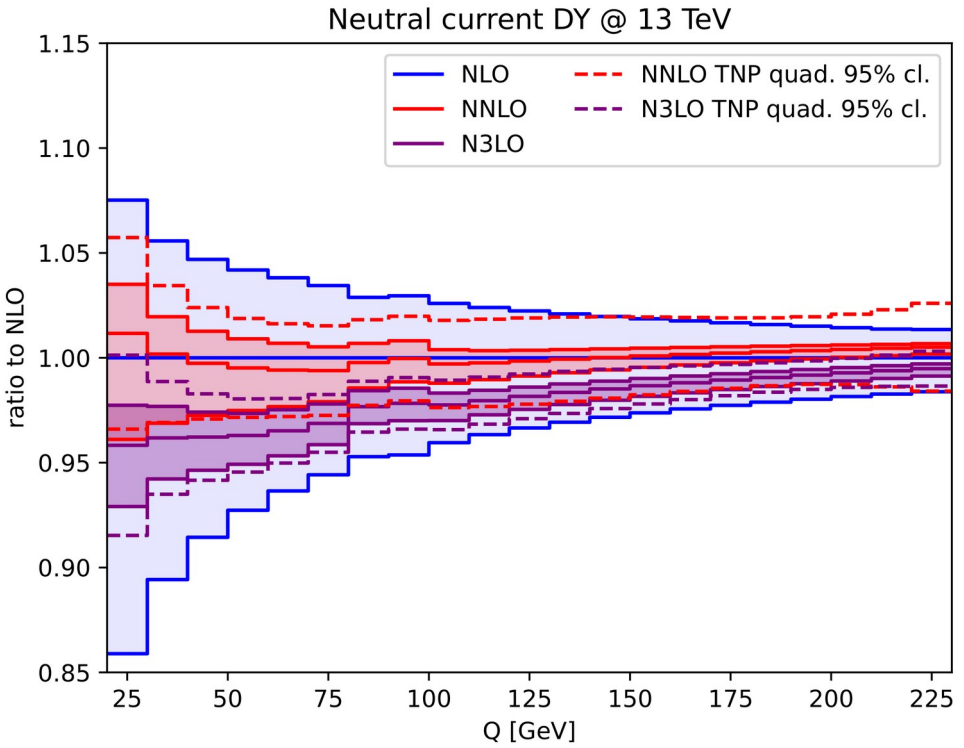
$$f_k^C\left(\vec{\theta}, x\right) = \frac{1}{2}\sum_{i=0}^{k} \theta_i T_i(x)$$



N3LO example: Drell-Yan

Numbers from n3loxs [Baglio, Duhr, Mistlberger, Szafron'22]





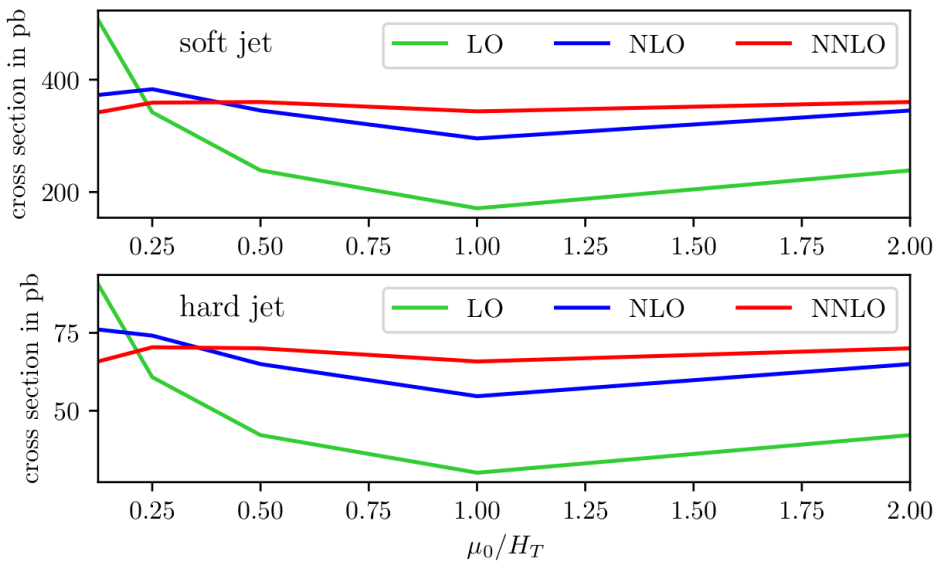
ttj: results for total cross section

LHC phase spaces at 13 TeV

Soft-jet: $p_T(j) > 30\,\mathrm{GeV},\ |y(j)| < 2.5$

anti-kT R = 0.4
PDF4LHC21

Hard-jet: $p_T(j) \geq 120\,\mathrm{GeV},\ p_T(j) \geq 0.4(p_T(t) + p_T(\bar{t})),\ |y(j)| < 2.4$



Dynamical scale choice:

$$\mu_R = \mu_F = \mu_0 = H_T/n$$

$$H_T = M_T(t) + M_T(\bar{t}) + p_T(j_1)$$

What I didn’t talk about

- Higher order EW corrections → important in the high energy tails
- Unfolding to stable top-quarks
  → highly non-trivial interplay between higher-order corrections in different phase spaces
- Parton-showers and matching?